

TOPIC 10. COMPOSITE MATERIALS

Processing of composite materials

- Polymer matrix composites (PMC)
- Metal matrix composites (MMC)
- Ceramic Matrix composites (CMC)
- Applications

Processing of Composites : PMCs

Laminating Procedures For Aircraft-Grade Composite Components:

Prep-peg production

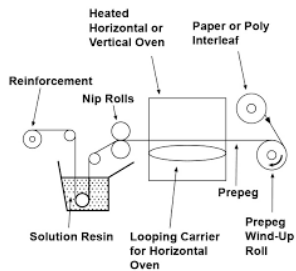
Prepeg transport and storage

Cutting and kitting

Lay-up or modling
(hand or automated)

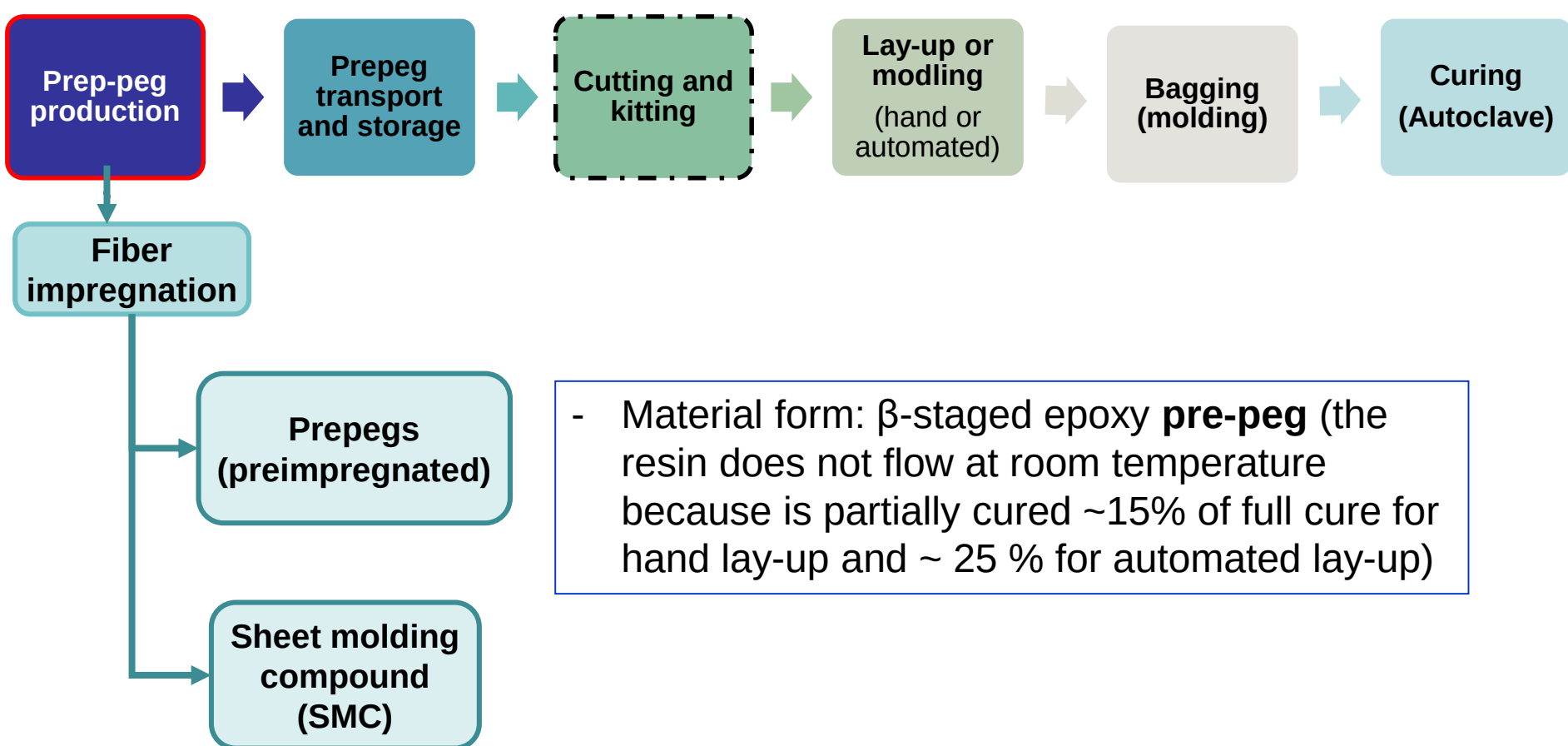
Bagging (molding)

Curing (Autoclave)



Processing of Composites : PMCs

Laminating Procedures For Aircraft-Grade Composite Components:



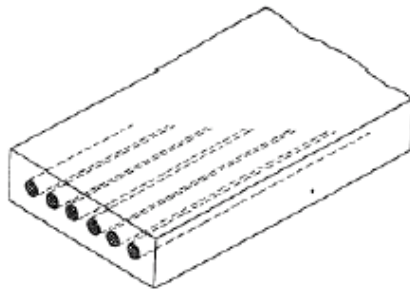
Processing of Composites : PMCs

What is a prepeg?

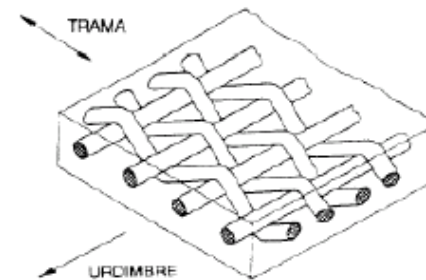
Resin-impregnated fiber forms (tape or fabric), commonly called **prepregs**, are manufactured by impregnating fibers with a controlled amount of resin (**thermoset or thermoplastic**), using solvent, hot-melt or powder-impregnation technologies.

Prepregs deliver a consistent fiber/resin combination and ensure complete wetout

Prepregs are the **most common way composite materials are used** in aeronautical technology



Unidirectional tape



Bi-directional fabric

- **Unidirectional tape (all fibers parallel)** is the most common **prepreg** form.
- **Prepregs made with woven fibers and other flat goods** offer reinforcement in two dimensions and are typically sold in full rolls, → provide three-dimensional reinforcement.

Processing of Composites : PMCs

What is a prepreg?

Prepregs with **thermosetting resins** can be stored in “ β -stage,” or partially cured state, until they are needed for fabrication:

- They require storage a temperature -18°C
- They have limited shelf life



Glass fiber / epoxy resin prepreg



Carbon fiber fabric / epoxy resin prepreg



Epoxy prepreg tape

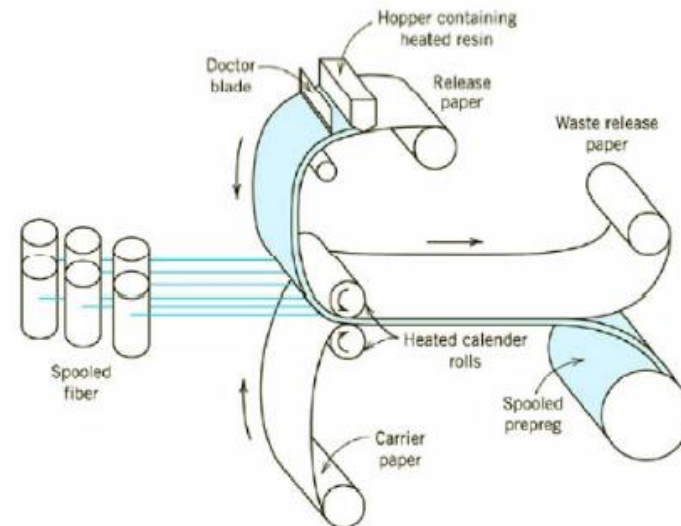
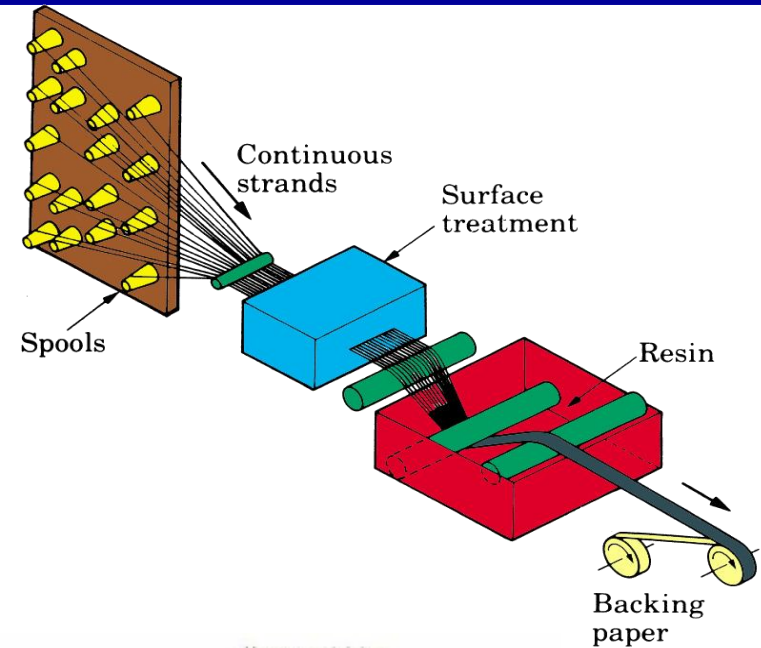
- Prepeg tape is assembled into laminated structures

Processing of Composites : PMCs

How are prepegs made?

Fiber Impregnation: prepeg

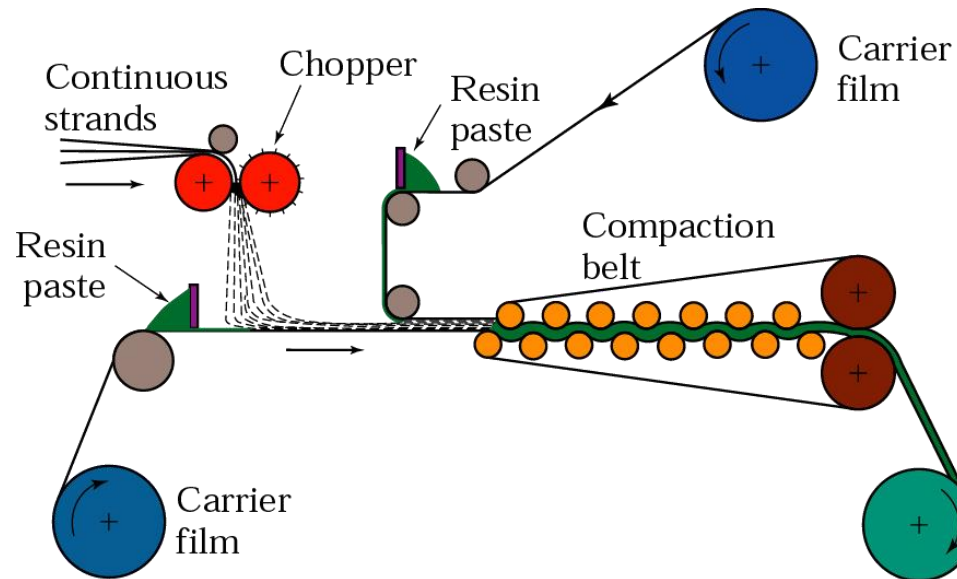
1. Continuous fibers are aligned and subjected to surface treatment
2. Fibers are coated by dipping in resin /solvent bath and a series of rollers ensure the correct amount of resin is impregnated
3. Tape is dried to evaporate the solvent and partially cured over cylinder.
4. The tape is protected by a backing paper and stored



Processing of Composites : PMCs

Fiber Impregnation: **sheet molding compound (SMC)**

1. Continuous fibers are chopped into short fibers
2. Short fibers are deposited over resin paste (random orientations)
3. Second layer of resin is deposited on top
4. Sheet is pressed between rollers

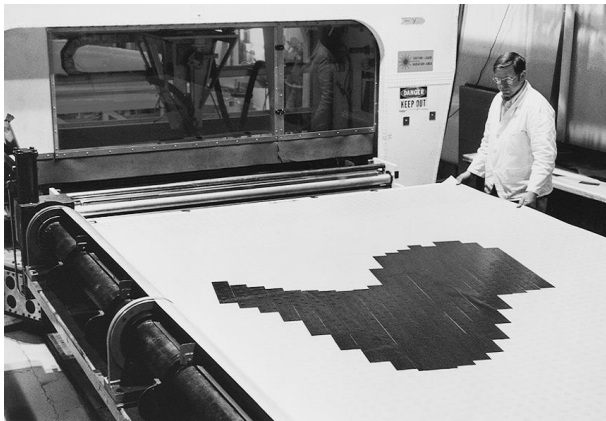


The sheet is still viscous at this stage; it can later be shaped into various products

*SMC is **not** as widely used in the aerospace industry as prepegs*

Processing of Composites : PMCs

Tape laying of prepegs into laminated structures

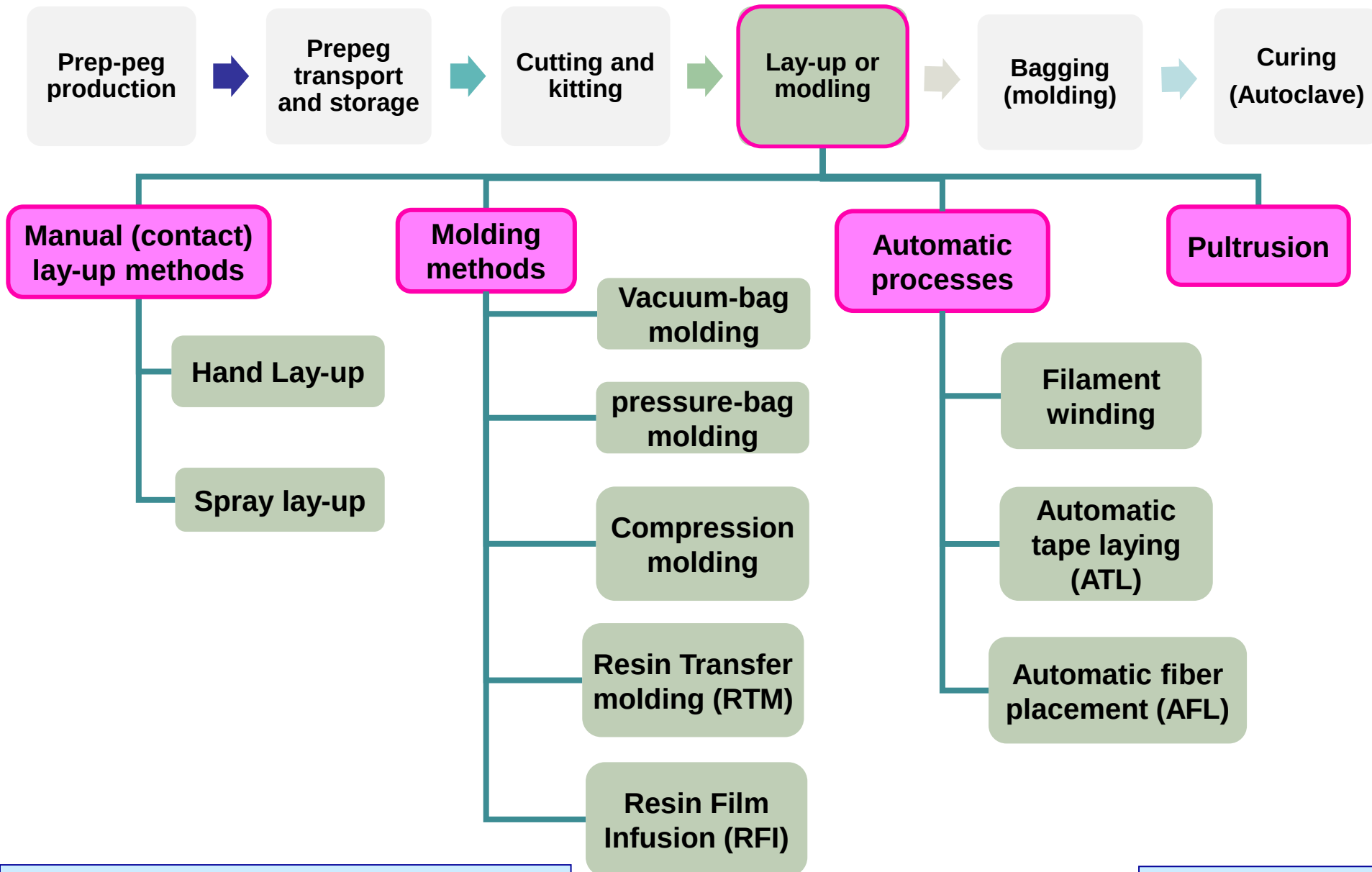


Single-ply lay-up of boron-epoxy tape for the horizontal stabilizer for F-14 fighter aircraft.



Automated prepeg cutting machine

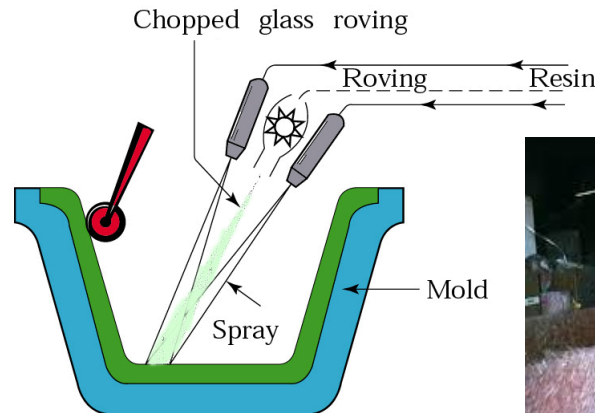
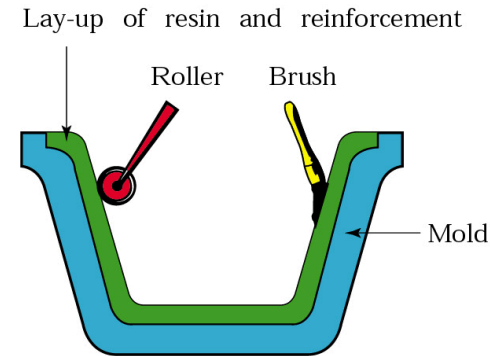
Processing of Composites : PMCs



Processing of Composites : PMCs

Manual (contact) lay-up methods :
used for *laminated* products or short
fibers

- Hand lay-up
 - Resins and reinforcements (prepregs) are placed in proper order and brushed or rolled into mold
- Spray lay up
 - Spray mixture of fibers + resin + curing agent into mold
 - Cannot use continuous fibers (short, randomly aligned fibers)



Curing is often required after molding (thermosetting polymers)

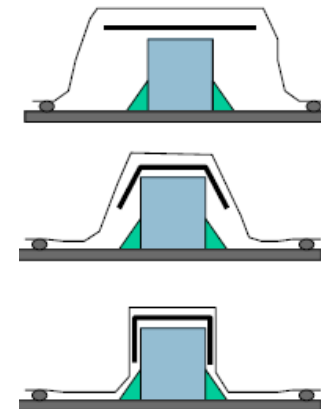
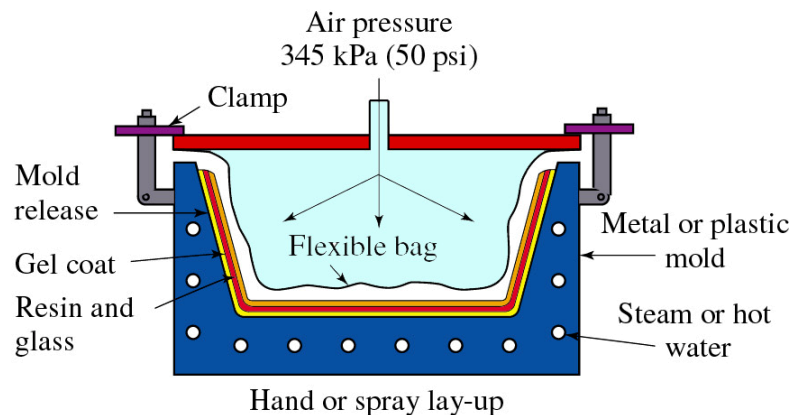
Processing of Composites : PMCs

Molding methods

- **Compression molding:**
 - Similar to polymer compression molding
 - Material is placed between two molds and pressure is applied
 - The material to be formed is a SMC or other fiber impregnated compound (i.e. bulk-forming compound: viscous mixture of polymer, fibers and additives)
 - The molds may be at RT or heated
- **Pressure-bag molding:**
 - flexible bag is placed over resin and reinforcing fiber mixture
 - Bag is pressurised



profile pressed in a mold

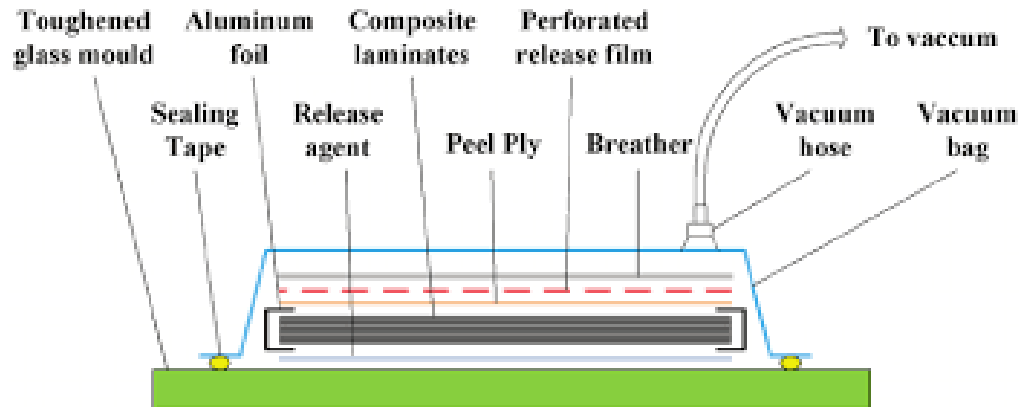
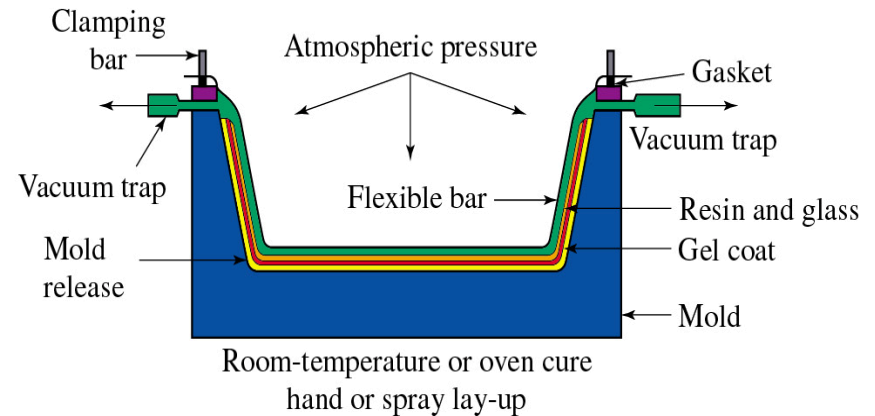


Processing of Composites : PMCs

Molding methods

- **Vacuum-bag molding:**

- Prepregs are laid in mold and covered with a bag or flexible bar
- Pressure is obtained as air is removed (vacuum)
- Vacuum maintained till curing is completed

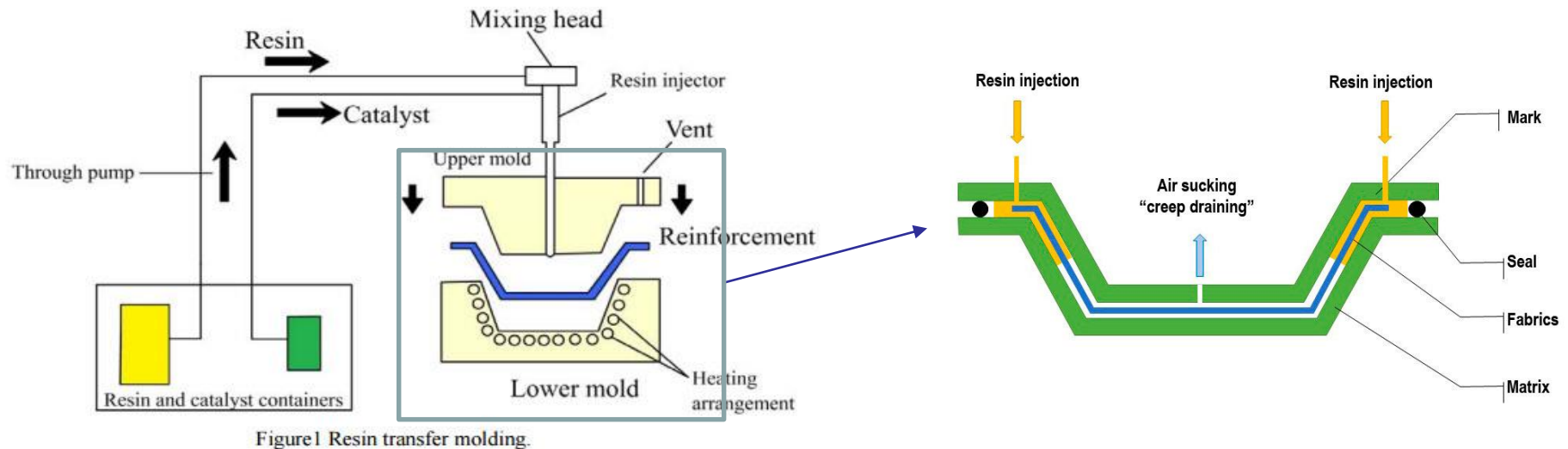


Curing is often required after molding (thermosetting polymers)

Processing of Composites : PMCs

Molding methods

- **Resin-Transfer molding (RTM) or Resin Injection Molding (RIM)**
 - Preform or fiber reinforcement (dry fabric) is placed into mold
 - Resin is forced into mold cavity (the mold is often in vacuum). Injection pressure ensure fiber impregnation
 - Curing



Uses **dry fabric** (non-impregnated) reinforcements or discontinuous reinforcement

Processing of Composites : PMCs

Molding methods

- **Resin-Transfer molding (RTM) or Resin Injection Molding (RIM)**

Most commonly used resins: epoxy and polyester



Control system for RTM



System ready for injection



Final part

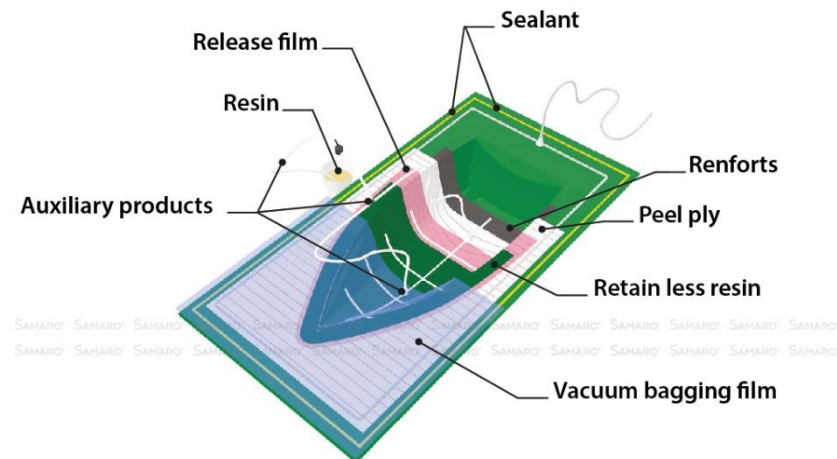
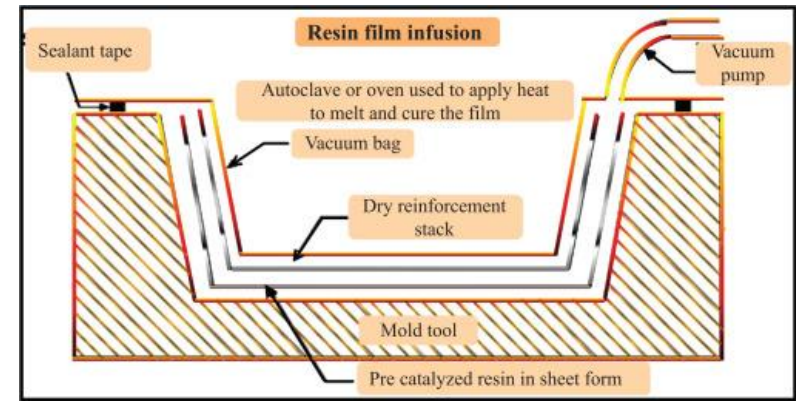
Can be used with thermoplastic matrices

Processing of Composites : PMCs

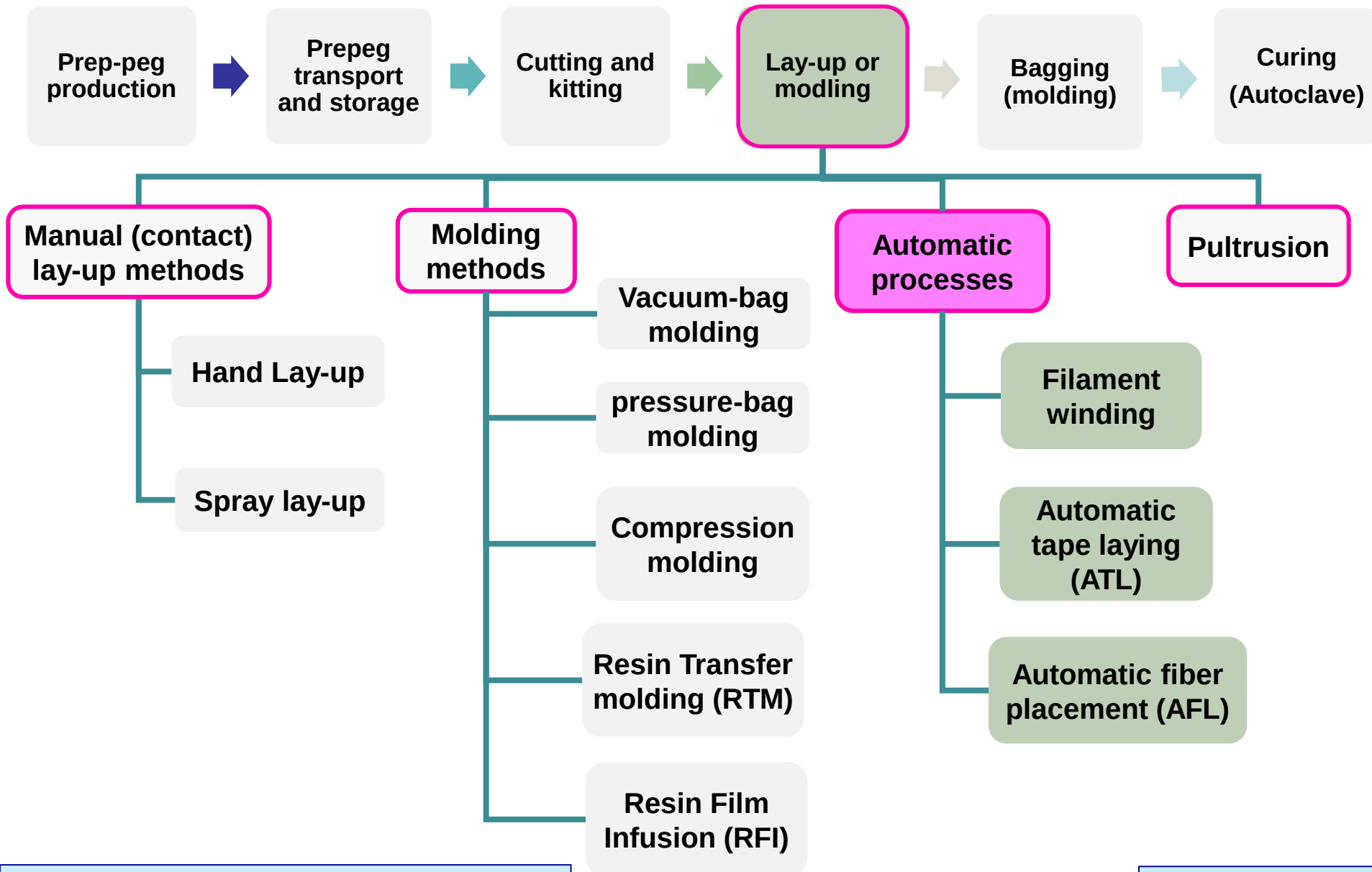
Molding methods

Resin Film Infusion, RFI

- Film shaped resin is placed in the mold along with the dry fiber.
- By applying heat and pressure the resin diffuses through the dry fiber to impregnate it thoroughly.
- This process can be performed in an autoclave, or simply by applying heat and vacuum.
- Can use both closed molds or open molds and vacuum bag.



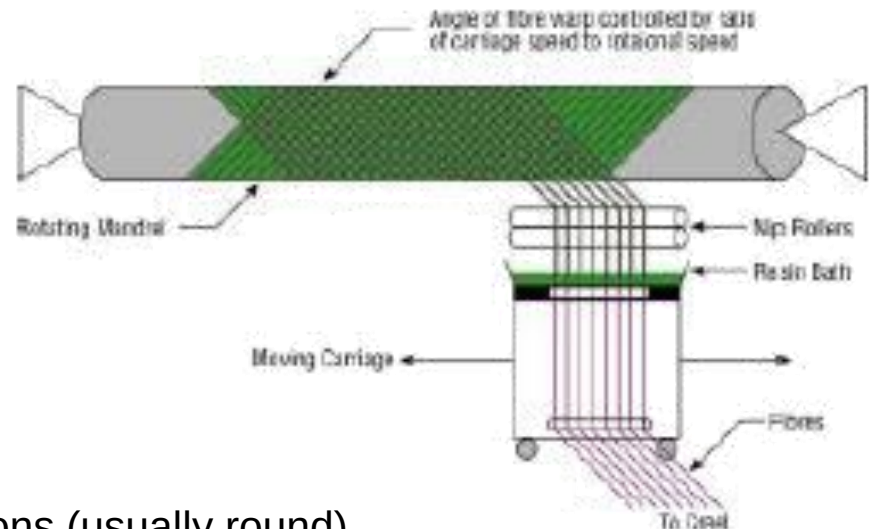
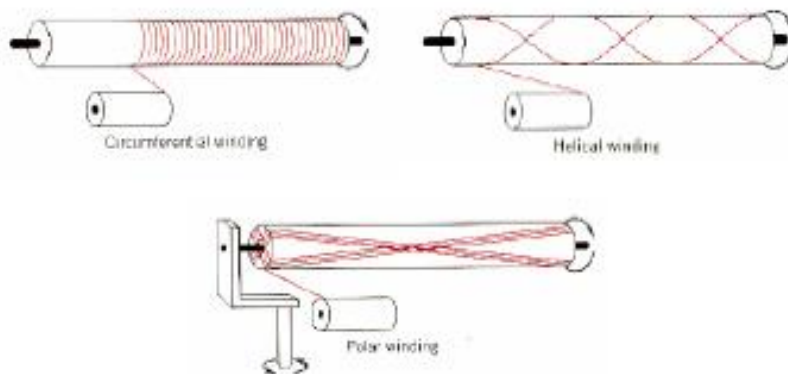
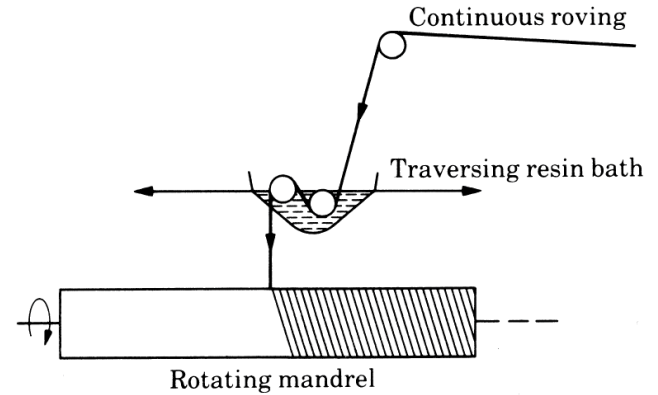
Processing of Composites : PMCs



Processing of Composites : PMCs

Filament Winding

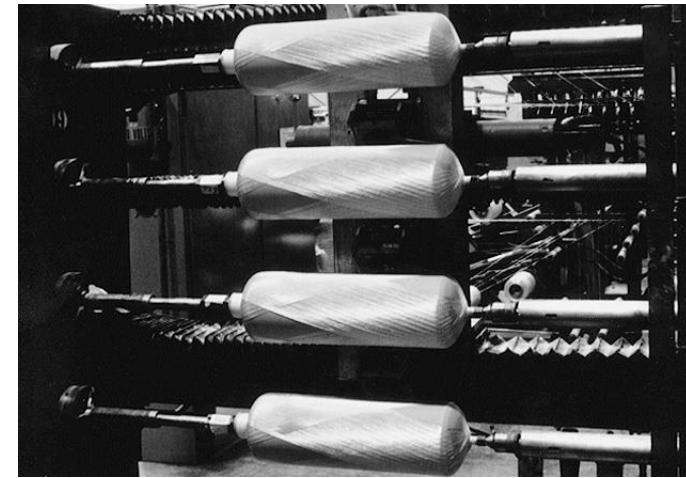
1. Reinforcement (filament, tape or roving) is impregnated by passing through polymer bath
2. Reinforcement is wrapped around a form
3. Resin and fibers are combined at the time of curing



- Form hollow parts with various cross-sections (usually round)
- Fibres, rovings, or narrow pre-pregs can be used.
- First automated process for producing fiber materials

Processing of Composites : PMCs

Filament Winding



Fiberglass being wound over aluminum liners, for slide-raft inflation vessels for the Boeing 767 aircraft.



Figure 15



Fiber Impregnation



Individual Tows

Figure 16

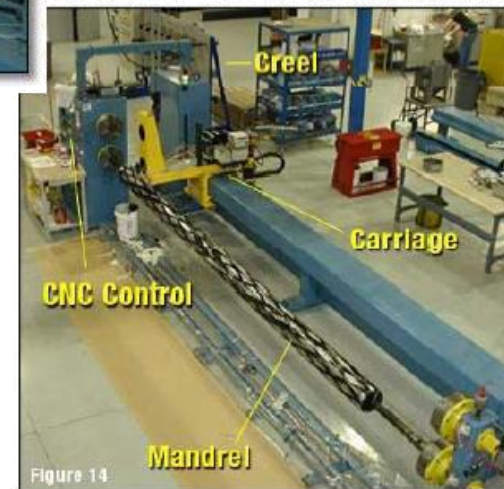
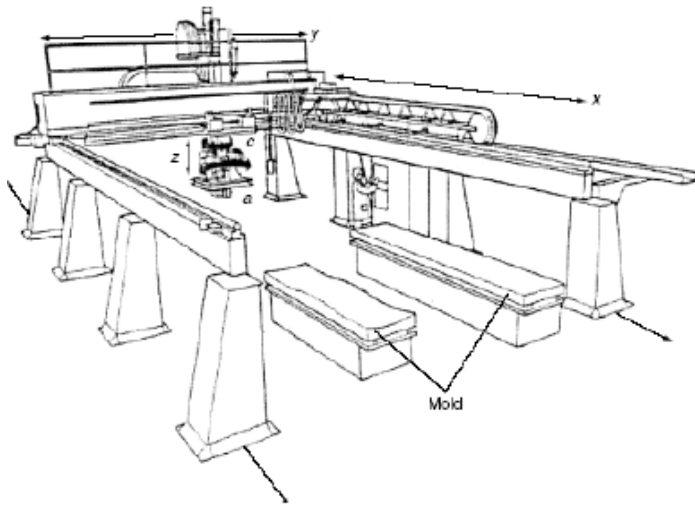


Figure 14

Processing of Composites : PMCs

Automatic tape laying (ATL)

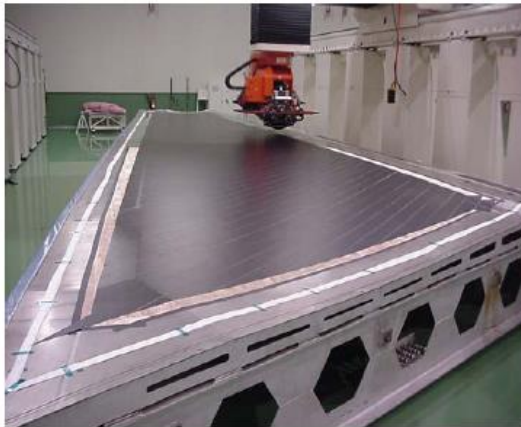
- Numeric control machines are able to put unidirectional tapes accuracy controlling both the orientation and the position
- Both on flat surfaces as slightly curved surfaces
- Usually unidirectional carbon fiber tape epoxy prepreg supplied with backing paper and with typical widths of 75, 150 and 300 mm
- The head carries the roll of tape, makes the necessary cuts and separates the plastic support (backing paper) as well as applying pressure on the mold.



Processing of Composites : PMCs

Automatic tape laying (ATL)

- Curvatures or slope changes in the mold, can cause changes in direction of the tape
- Typical application: manufacture of multilayer laminated tape superimposed with different orientation angles of the reinforcement. The panel is cut and the cut pieces are stacked and hot formed to be structural parts such as spars, ribs etc.



Automatic tape laying machine



A 10 axis computer-numerical-controlled tape-laying system. This machine is capable of laying up 75 mm and 150 mm (3 in. and 6 in.) wide tapes, on contours of up to $\pm 30^\circ$ and at speeds of up to 0.5 m/s (1.7 ft/s)

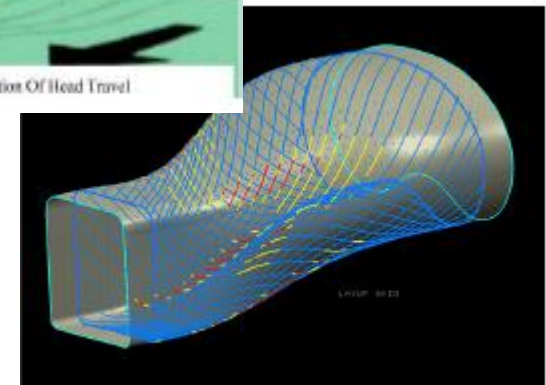
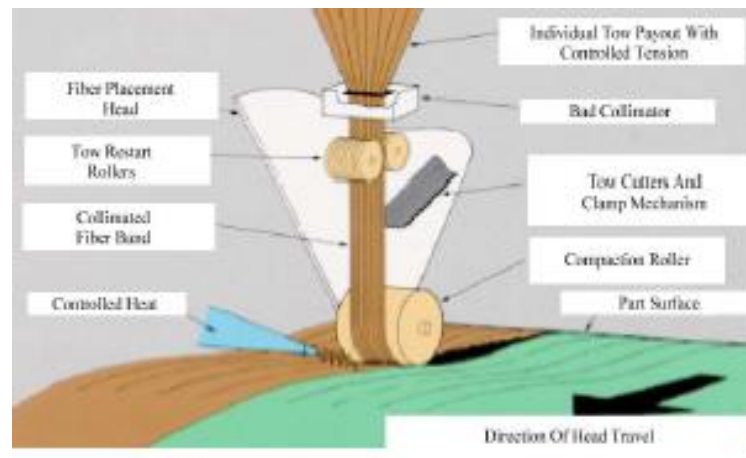
Processing of Composites : PMCs

Automatic Fiber Placement (AFP)

- Allows automating composite lamination. Combines the advantages of filament winding and automatic tape laying, solving the limitations of both methods.
- Can automatically lay up both concave and convex surfaces, closed or open.
- The material is deposited without tension on the tool surface. It can be cut and re-fed automatically.
- Capable of controlling a large number of narrow-width tapes or tows (3 to 6 mm) instead of a single tape.



Fiber Placement machine

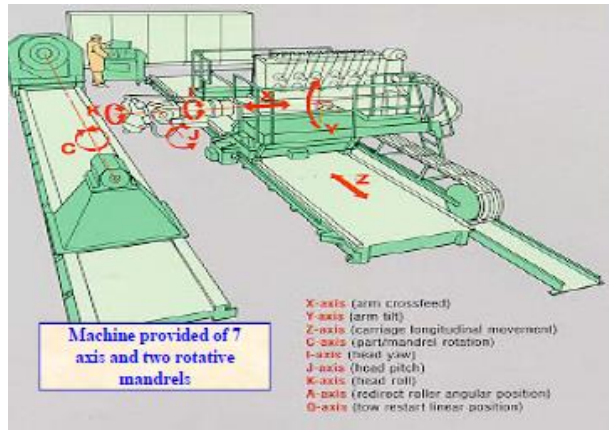


Fiber Placement Interface software

Depositon rate is lower that ATL

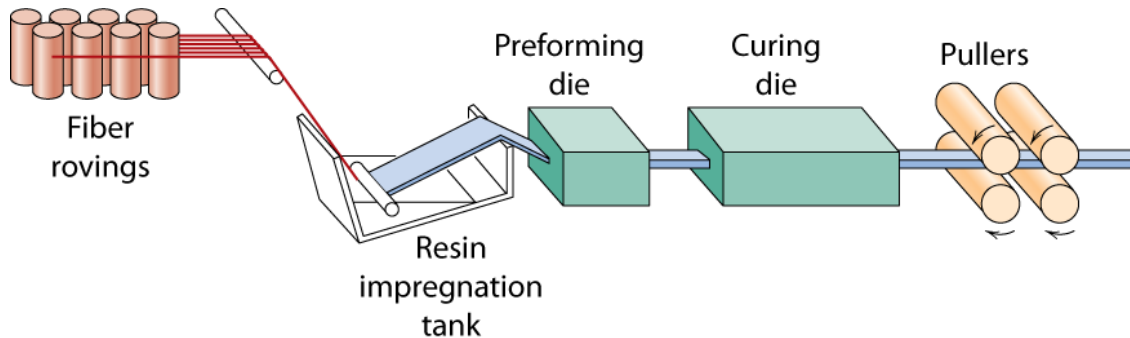
Processing of Composites : PMCs

Automatic Fiber Placement (AFP)



Processing of Composites : PMCs

Pultrusion



1. Roving (continuous fibres) are impregnated with resin (matrix).
2. Preform die establishes shape of cross-section and fibre/matrix ratio
3. Curing die is heated to accelerate cure of matrix (Thermosetting).

- Used for shapes with constant cross section: rods, tubes, beams.
- Fibres: Glass, carbon, aramid.
- Matrix: Polyester, vinyl esters, epoxy



Processing of Composites : PMCs

Laminating Procedures For Aircraft-Grade Composite Components:

Prep-peg production



Prepeg transport and storage



Cutting and kitting



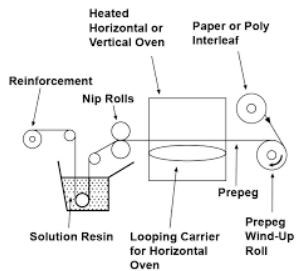
Lay-up or modling
(hand or automated)



Bagging (molding)



Curing (Autoclave)



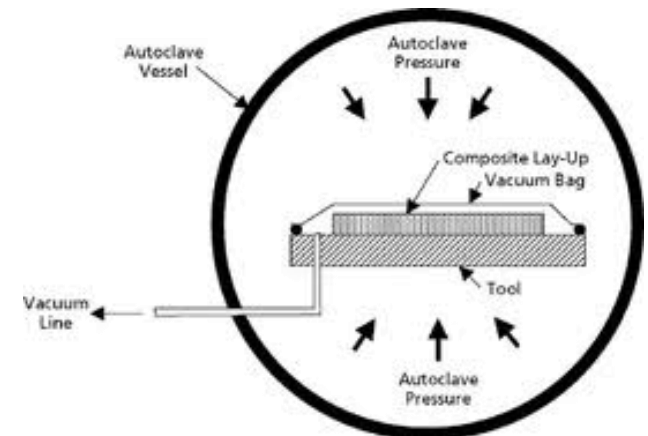
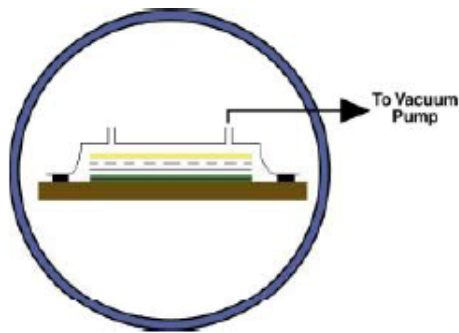
Processing of Composites : PMCs

Curing: in some cases high pressures are need in order to obtain adequate curing of complex pieces. Curing is of PMC with thermosetting matrix is often performed in an *autoclave*.

Autoclave: a very large, internally heated pressure vessel, with internal connections for vacuum hoses and sensors such as thermocouples



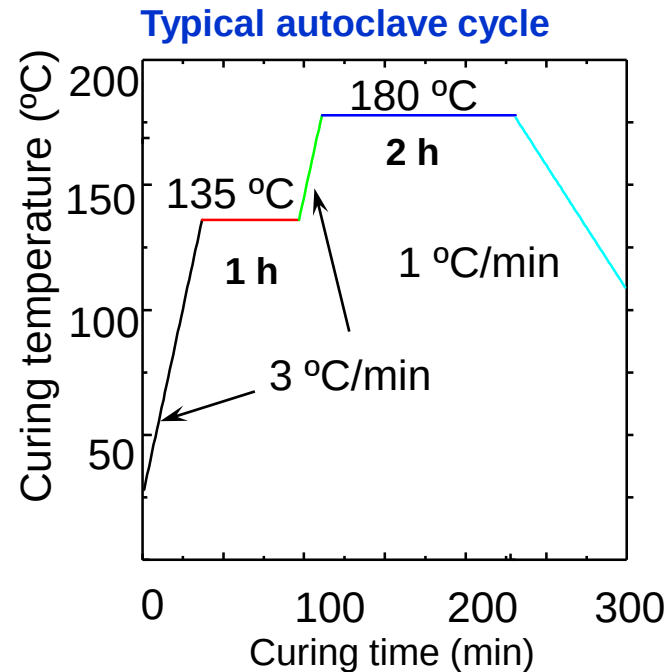
Composite curing autoclave



Processing of Composites : PMCs

Curing (Autoclave):

- **First step**: viscosity reduction, voids and air bubbles elimination, pressure is applied
- **Second step**: curing, temperature is raised to T_c (usually 120° or 180°C for thermosetting matrices)



To best results: pressure above 1 atm

Processing of Composites : PMCs

Integration of structures

The processes described are capable of producing relatively simple parts that **need to be integrated** to reduce the cost of subsequent assembly processes.

- In the **co-curing** process, a number of simple *uncured parts* are assembled using a set of tools along with the vacuum. This assembly is introduced into the autoclave, which produces the polymerization of the resin.
- By **bonding** processes, several simple *cured* parts and, are positioned and stuck using *adhesives* which cure at high temperature and pressure.
- **Co-bonding** is the process to join the various simple parts with *adhesive*, some of which are *pre-cured* with others that are *not cured*. The polymerization of the resin and adhesive occur simultaneously.

For complex and highly optimized weight structures, more level of integration requires a more complex tooling and therefore is more costly. The cost optimization can be achieved with simplified structures, whose integration can be done with more flexible tooling

Processing of Composites : PMCs

Integration of structures: Example



Processing of Composites : PMCs

Integration of structures



Co-bonded pre-cured skin integrated con wet T-stringers meanwhile turning rack

METAL MATRIX COMPOSITES

Common **Matrices**: Al, Mg, Ti.

Light metals are used to form high-performance metal-matrix composites

Advantages:

- higher **specific strength** and stiffness
- Higher service **temperature capabilities** than PMCs.
- greater **wear resistance**
- **tailored properties** for a particular application

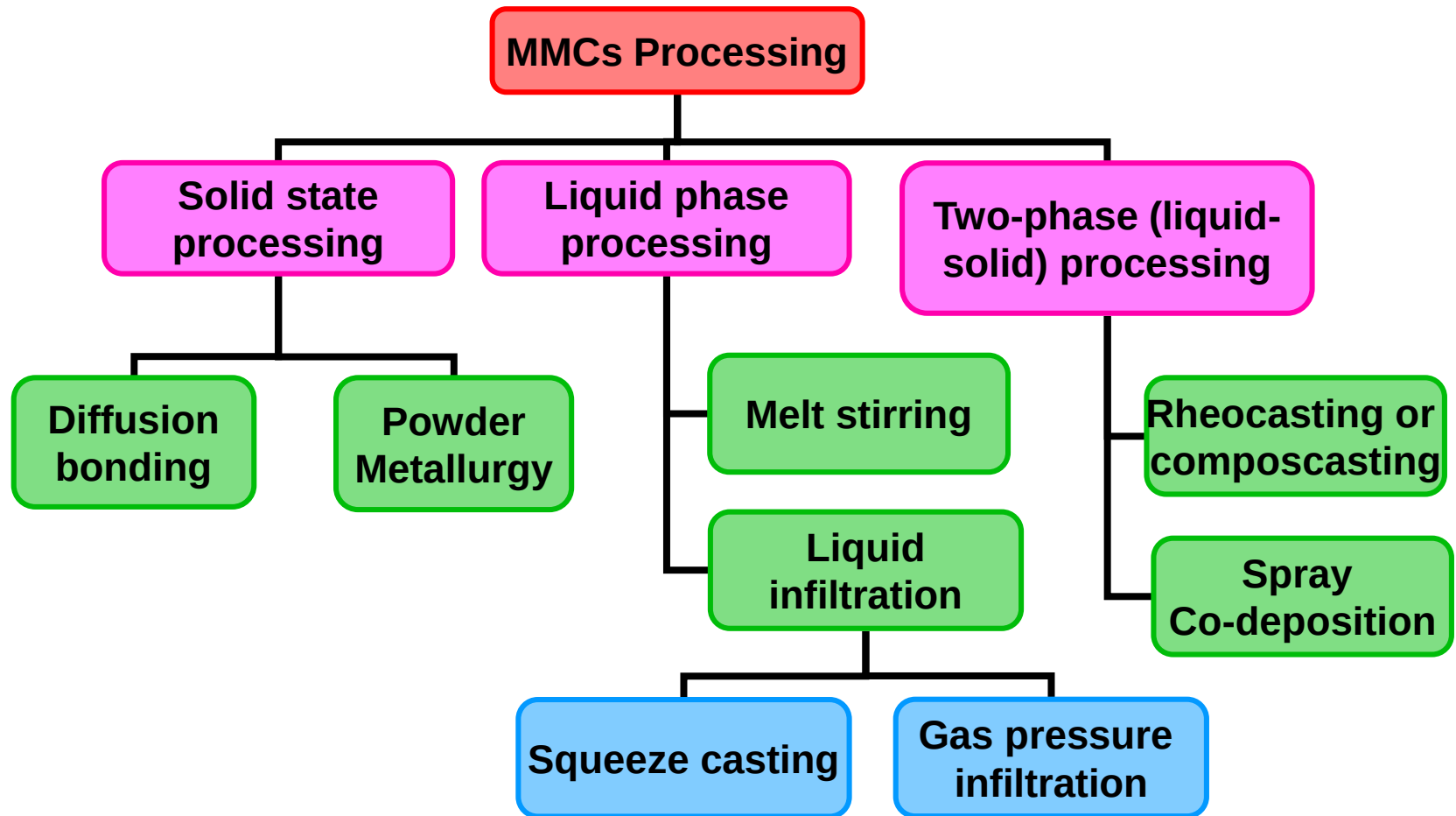
Common reinforcements:

- Ceramic reinforcements : SiC, Al₂O₃ and Borsic (boron fibres coated with silicon carbide).
- Carbon fibres may be used with Al or Mg matrices.

Disadvantages:

- Costly, complex and limited fabrication techniques.
- Metals often **react chemically** with and weaken fibers during manufacture or in service at elevated temperatures.

Processing of Composites : MMCs



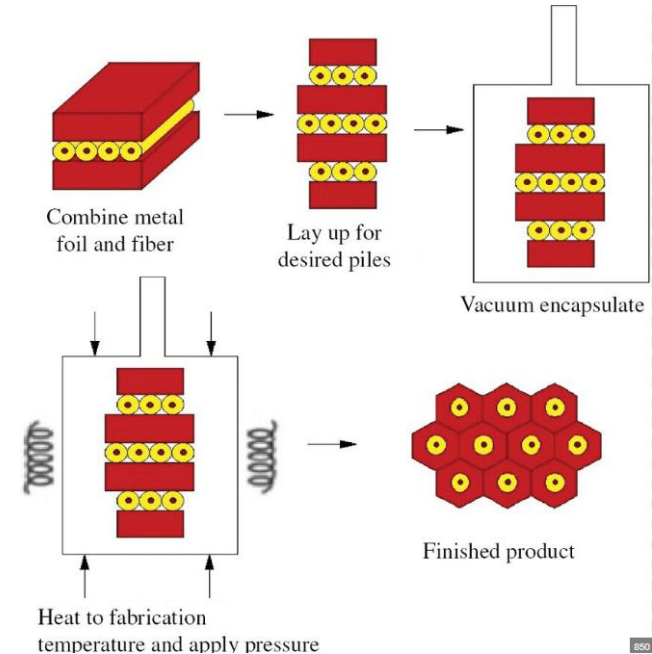
Processing of Composites : MMCs

Solid state processing

Diffusion bonding

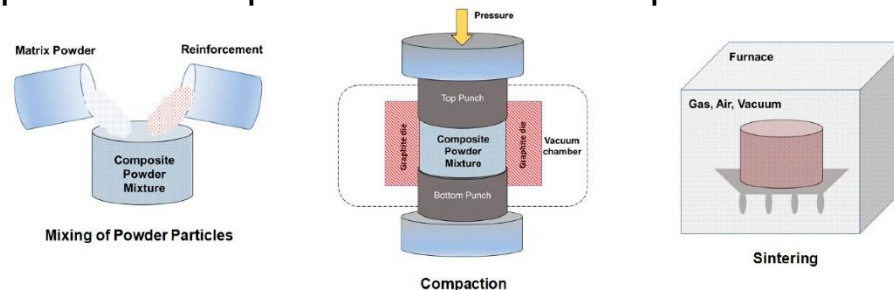
- Sandwich fiber mat between two sheets of foil to form a *ply*
 - Consolidate (by heat process)- not essential
 - Cut and stack plies
 - Hot press
- Expensive and limited to simple shapes

<http://www.youtube.com/watch?v=5EhMYMr834o>



Powder metallurgy

- Mix components and process in traditional powder metallurgy route



- *Suitable for discontinuous fiber, whiskers or particulates*

Processing of Composites : MMCs

Liquid phase processing

Melt stirring

Mix reinforcement into molten metal and cast

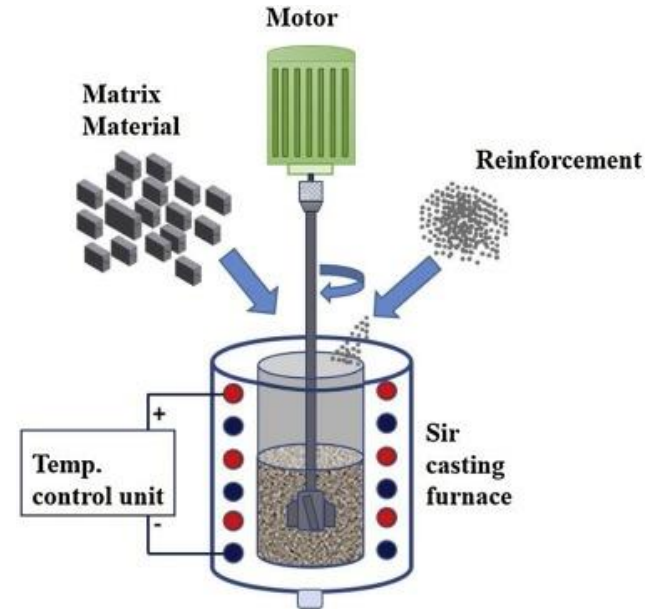
Limited process. Mixing is poor even with stirring due to density differences

Liquid infiltration

Liquid forced **into porous reinforcement preform** by applying pressure

Gas pressure infiltration

Squeeze casting



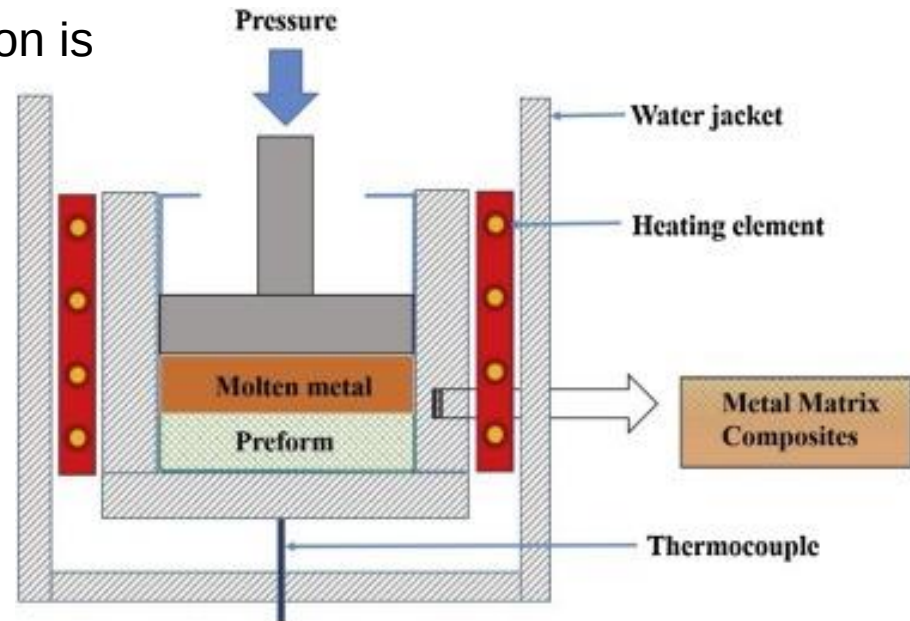
Processing of Composites : MMCs

Liquid phase processing

Liquid infiltration

Squeeze casting

- Insert porous preform into die
- Add correct amount of molten metal
- Apply pressure mechanically
- Release pressure when solidification is complete
- Remove from die



Suitable for small near net shape products

Processing of Composites : MMCs

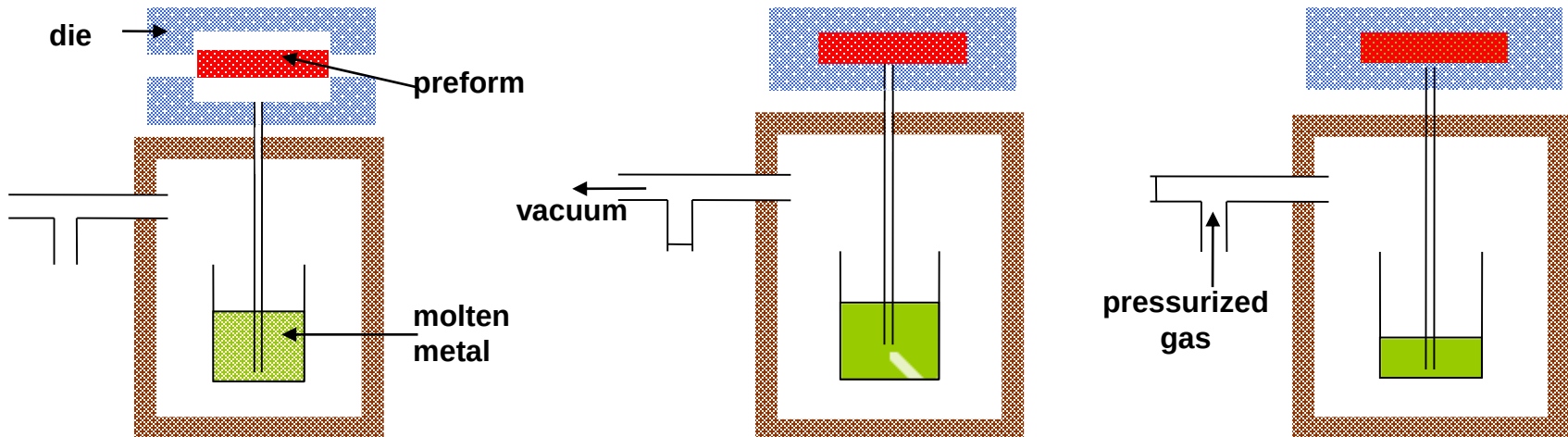
Liquid phase processing

Liquid infiltration

Gas pressure infiltration

- Preform is inserted into closed die
- Chamber containing molten metal is evacuated
- Gas is forced into chamber under pressure
- Pressure causes molten metal to infiltrate the preform
- Pressure is maintained until solidification is completed
- Remove from die

Suitable for larger components

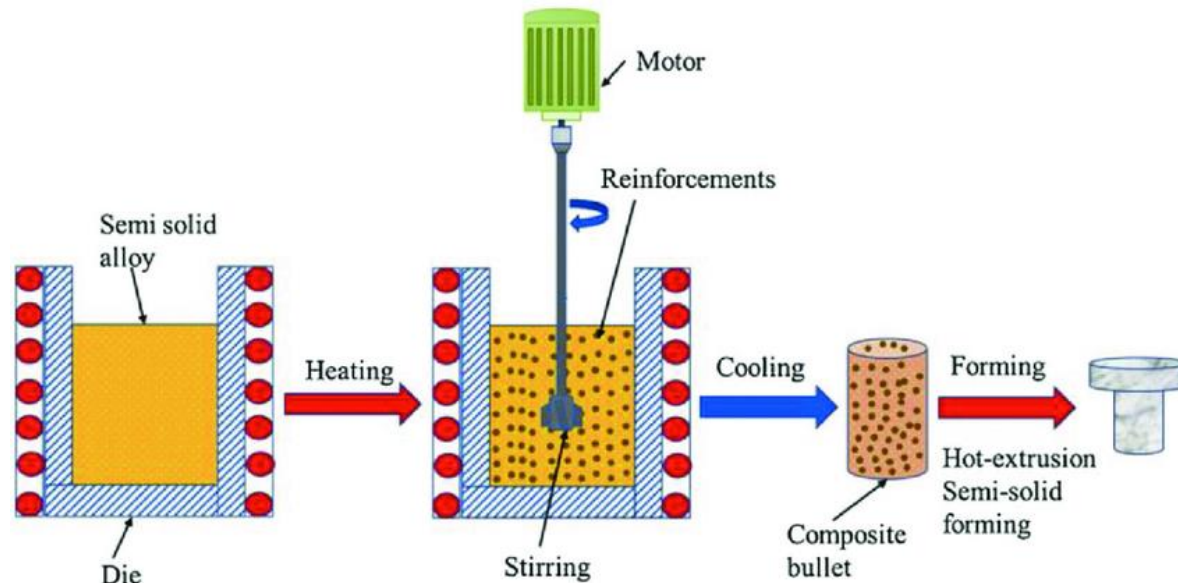


Processing of Composites : MMCs

Two-phase (liquid-solid) processing

Rheocasting or composcasting

- Modification of melt-stirring (to have better mixing)
- Mix reinforcement with metal when metal is in mushy two-phase region liquid-solid state



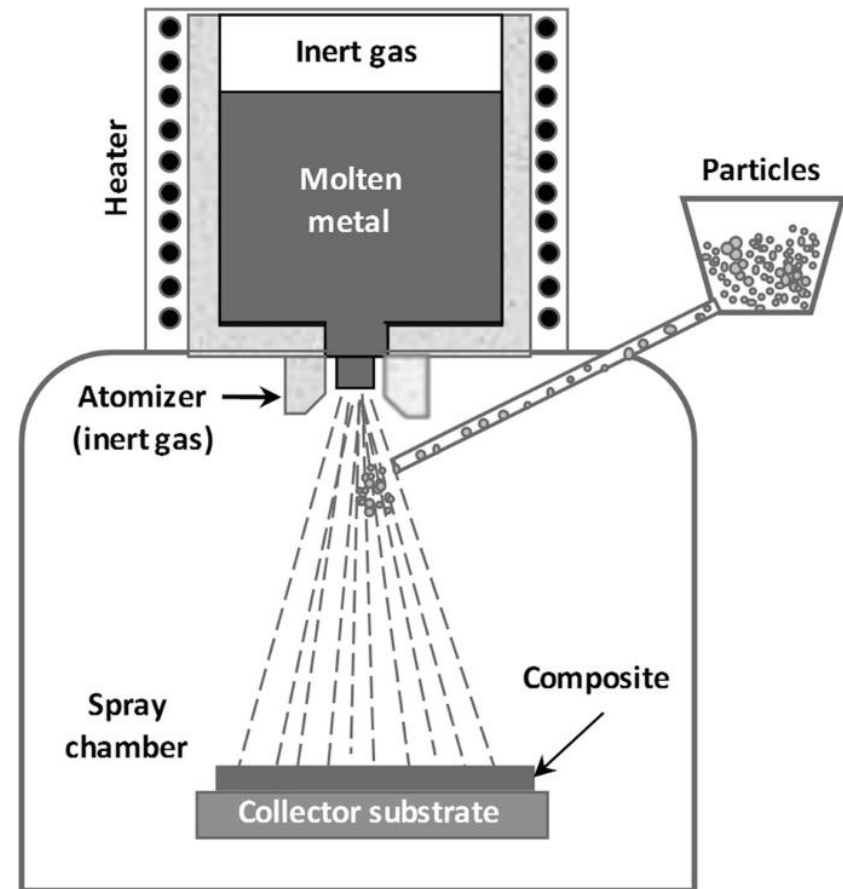
Limited to casting alloys with wide solidification range

Processing of Composites : MMCs

Two-phase (liquid-solid) processing

Spray Co-deposition

- Atomize the melt
- Introduce reinforcement into spray of melt droplets
- Co-deposit onto substrate



Processing of Composites : CMCs

CERAMIC MATRIX COMPOSITES

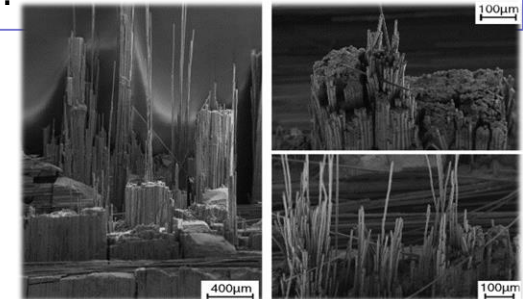
Common **Matrices**: silica based glass, silicon carbide, silicon nitride or alumina or a carbon matrix.

Advantages:

- High Fracture toughness or crack resistance
- Low density
- High specific strength and specific modulus
- High temperature oxidation stability.
- Greater **wear resistance**
- Extreme conditions

Common reinforcements:

- Ceramic reinforcements: **SiC**, **Al₂O₃** and **Borsic** (boron fibres coated with silicon carbide).
- **Carbon fibres** may be used
- Purpose **increase toughness** → increase fracture energy through various **energy-absorbing mechanisms** (crack bridging or fibre pull-out).



Fracture surface of a fiber-reinforced ceramic composed of SiC fibers and SiC matrix

Disadvantages:

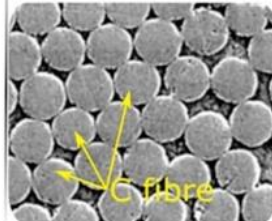
- Costly, complex and limited fabrication techniques.

Processing of Composites : CMCs

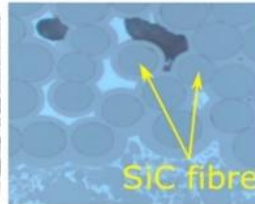
CERAMIC MATRIX COMPOSITES Examples

Silicon Carbide
matrix composites**Key points**

Used in high thermo-structural applications with long time exposures.

Reinforcements:

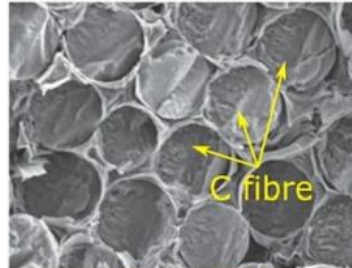
Heidenreich, 2015



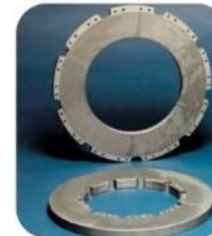
Gavalda et al., 2018

Applications:Rocket nozzle
Di Carlo, 2015Aeroengine vane
Spriet, 2015Carbon
matrix composites**Key points**

Used in frictional applications and short high temperature applications.

Reinforcements:

Hatta et al., 2015

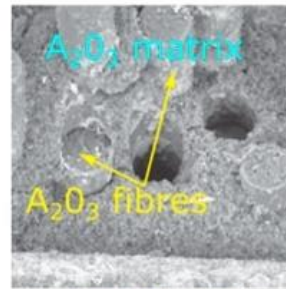
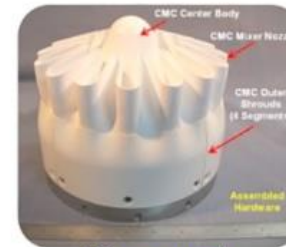
Applications:Break discs
Krenkel et Thebault, 2015Ventilators gas
Hatta et al., 2015

Processing of Composites : CMCs

CERAMIC MATRIX COMPOSITES Examples

Oxide
matrix composites**Key points**

Used in applications where superior thermal stability is needed.

Reinforcements:Keller et al., 2015 10 μm **Applications:**

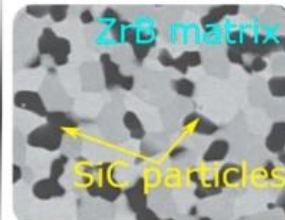
Mixer nozzle
Keller et al., 2015



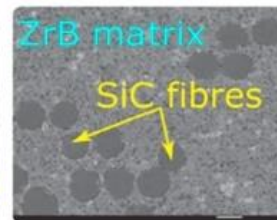
Acoustic nozzle
Keller et al., 2015

 ZrB_2 or HfB_2
matrix composites**Key points**

Used in applications where ultrahigh temperature is needed ($>2000^\circ\text{C}$)

Reinforcements:

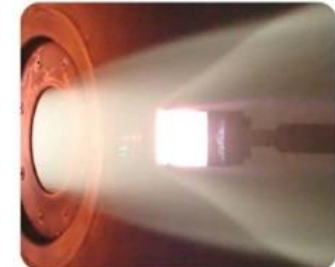
Kagawa et al., 2015



Kagawa et al., 2015

Applications:

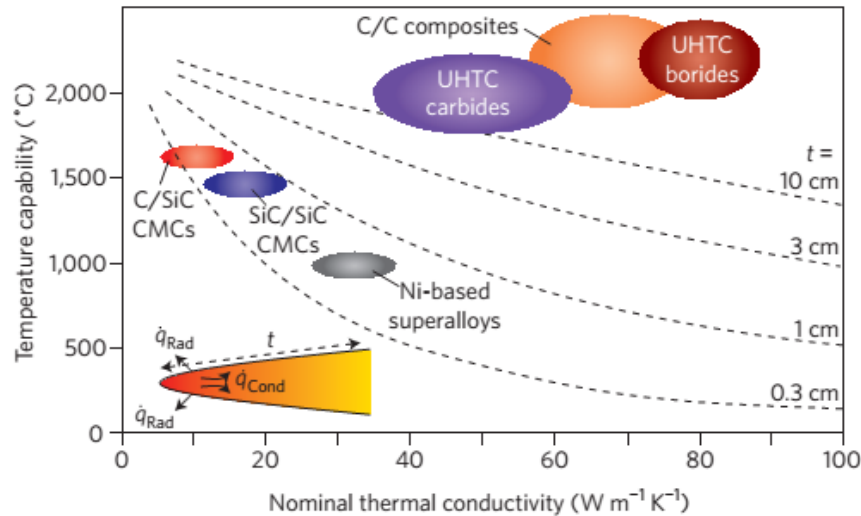
Rentry nose
Johnson, 2015



Nozzle
Savino et al., 2015

Processing of Composites : CMCs

CERAMIC MATRIX COMPOSITES Examples



An F-16 Fighting Falcon F100 engine exhaust nozzle with five A500 Ceramic Matrix Composite divergent seals, identified by the yellow arrows.



Fig. 9. Vulcain subscale nozzle during 40 bar hot-firing test (F3).



Fig. 7. Vulcain subscale C/SiC nozzle extensions (1:5).

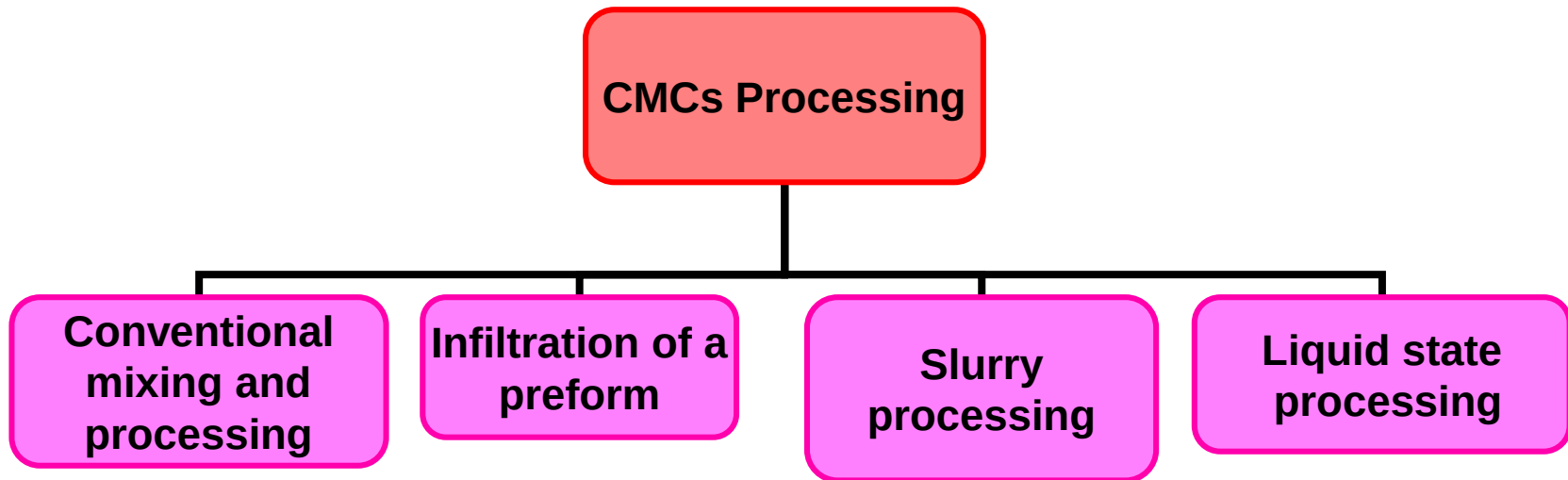
Carbon reinforced with silicon carbide (C/SiC)

Silicon Carbide reinforced with silicon carbide (C/SiC)

C-C composites

<http://www.wpafb.af.mil/news/story.asp?id=123105651>

Processing of Composites : CMCs

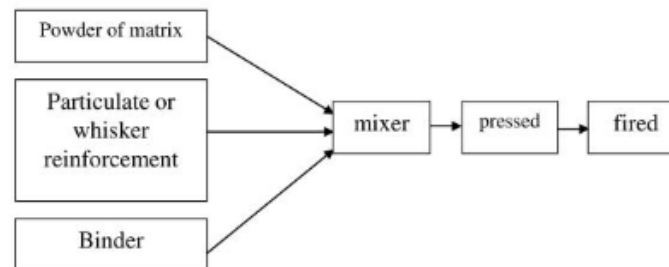


Processing of Composites : CMCs

Liquid state processing

- Molten matrix constituent is infiltrated between reinforcement phase under pressure (used widely in MMC)
- Limited success in CMCs due to:
 - High melting temperature of ceramics causes reaction between matrix and reinforcement
 - High viscosity of ceramics causes low infiltration rate

Conventional mixing and processing

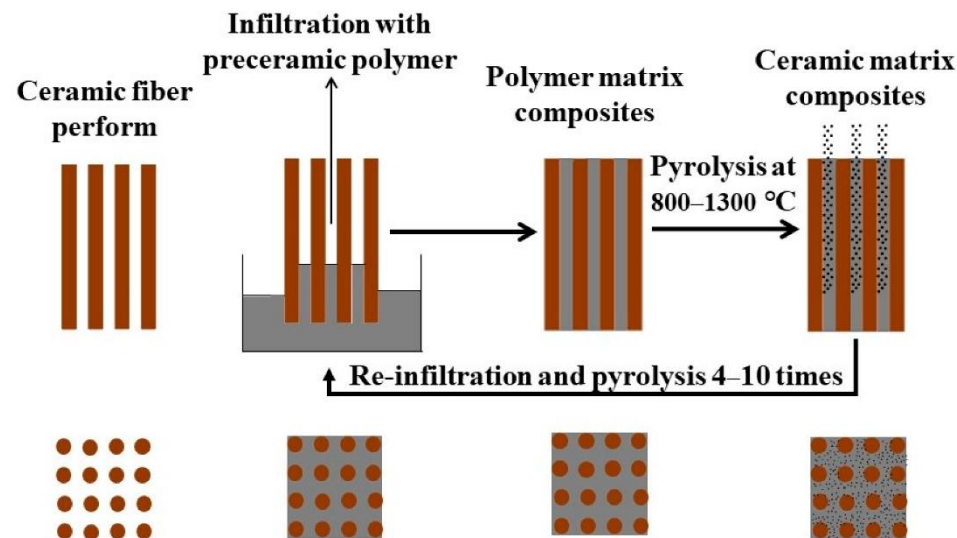
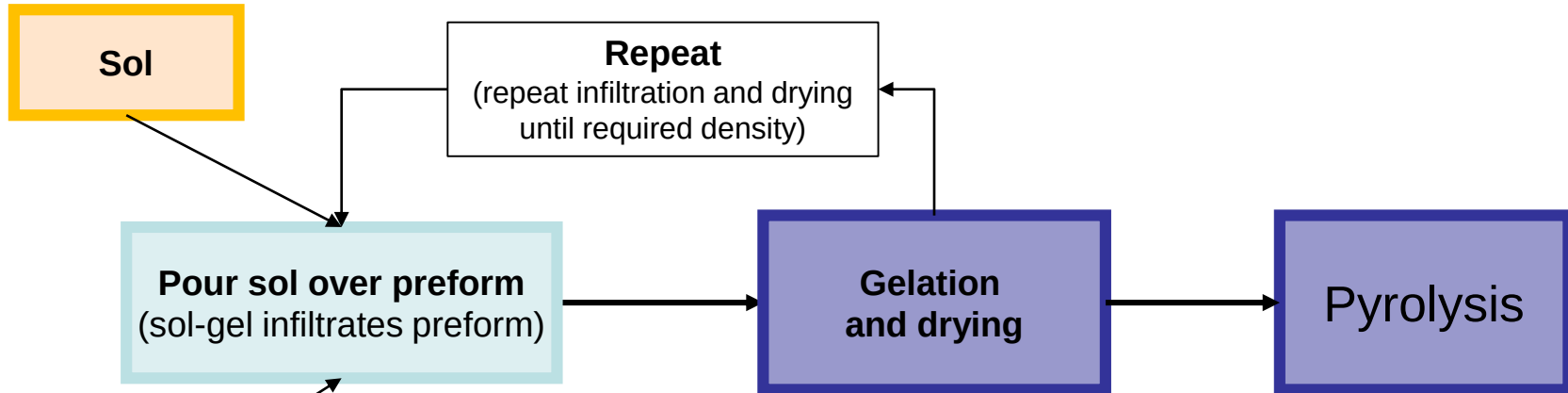


Problems:

1. nonuniform mixing
2. low volume fraction of reinforcement
3. damage of whiskers during mixing and pressing

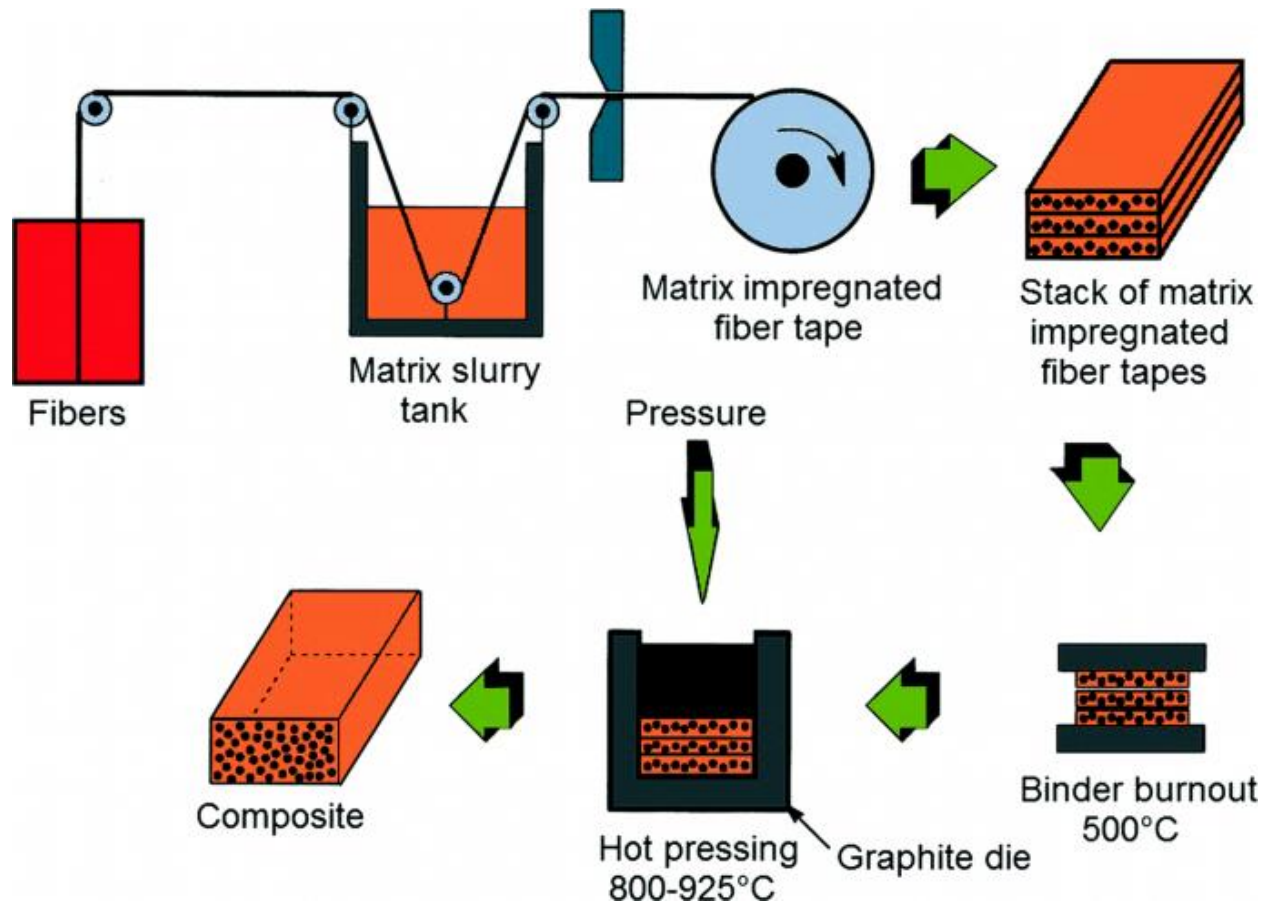
Processing of Composites : CMCs

Infiltration of a preform



Processing of Composites : CMCs

Slurry processing



Applications

- ▶ Aerospace Industry
- ▶ Military applications
- ▶ Naval Industry
- ▶ Transportation
- ▶ Construction Industry
- ▶ Chemical Industry
- ▶ Sports equipment
- ▶ Medical Materials
- ▶ Consumer goods

Material Requirements

- ▶ High specific strength
- ▶ Low coefficient of thermal expansion
Extreme temperature changes (-200 a 200 °C)
- ▶ Dimensional stability
Critical parameter for some applications
- ▶ Low weight
Critical for special application

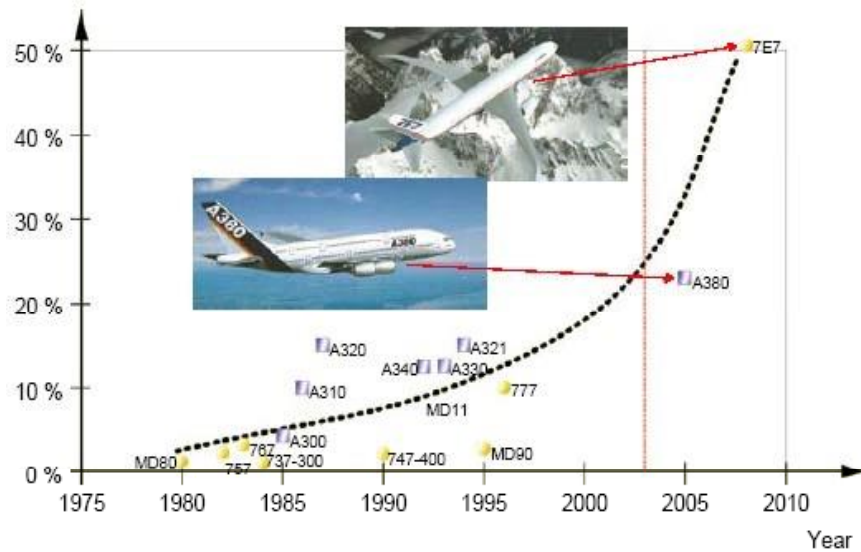
The **new challenges** in composite materials for aircraft :

- ☐ Toughened thermosetting polymers
- ☐ High-performance thermoplastics
- ☐ Innovative low-cost manufacturing processes
- ☐ Recyclability

Composites materials in aircraft structures

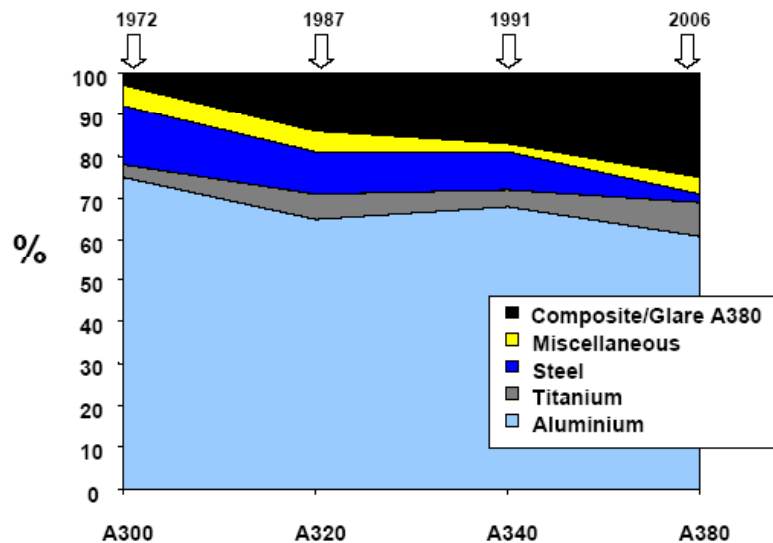
Aircraft Model	Component	Material	Number of Parts	Entered Service
L-1011	Underwing fairing	Aramid/epoxy	18	1973
L-1011	Aileron	Carbon/epoxy	8	1982
DC-10	Upper aft rudder	Carbon/epoxy	15	1976
DC-10	Aft pylon skin	Boron/aluminum	3	1975
DC-10	Vertical stabilizer	Carbon/epoxy	1	1987
727	Elevator	Carbon/epoxy	10	1980
737	Spoiler	Carbon/epoxy	108	1973
737	Horizontal stabilizer	Carbon/epoxy	10	1984

Share of Composite Components



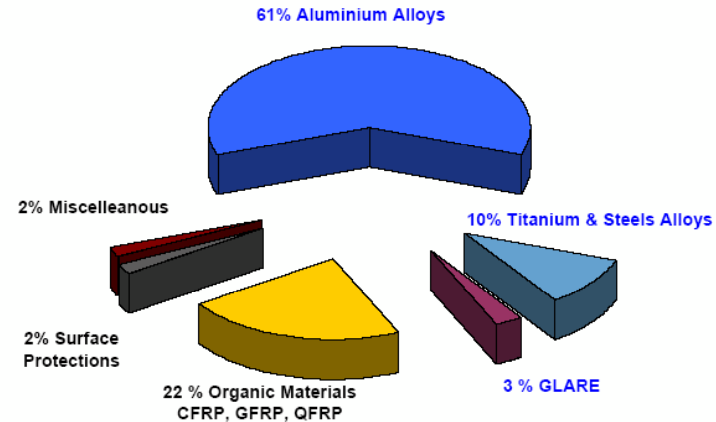
- mid-**1960s**: Glass-reinforced composites (honeycomb sandwich constructions **secondary structures** (i.e., wing-to-body fairings, fixed-wing and empennage cover panels, and **secondary control surfaces**))
- early 1980s: Carbon/epoxy, aramid/epoxy, and aramid-carbon/epoxy and glass-carbon/epoxy hybrid composites used in secondary structures such as belly **fairings**, **fixed-wing** and **empennage cover panels**, and **engine cowlings**, as well as primary flight controls such as **ailerons**, **elevators**, **rudders**, and **spoilers** (Boeing 757, 767, and 737-300; Airbus A310 and A320)

Composites materials in aircraft structures



Material Distribution in Airbus Aircraft

Structure Material Breakdown
(Engine, Landing gear not included)



A380-800 Materials Distribution

- late 1980s : application of composites on primary structure **empennage components** (Airbus A320) . **carbon/epoxy laminate skin** fabricated from materials similar to the first-generation materials previously used for secondary structure and primary flight controls.
- Development of **carbon fibers with improved strength and modulus** (e.g., Hercules IM7 and Toray T- 800H) and **high-performance and toughened matrix polymers** led to application on the Boeing 777 **empennage**

Composites materials in aircraft structures

A340-600 has approximately 15 % of the structural weight consists of carbon fibre reinforced composites.



Wing J-Nose



Engine Cowling

Vertical Tail Plane

Horizontal Tail Plane



Keel Beam



Belly Fairing

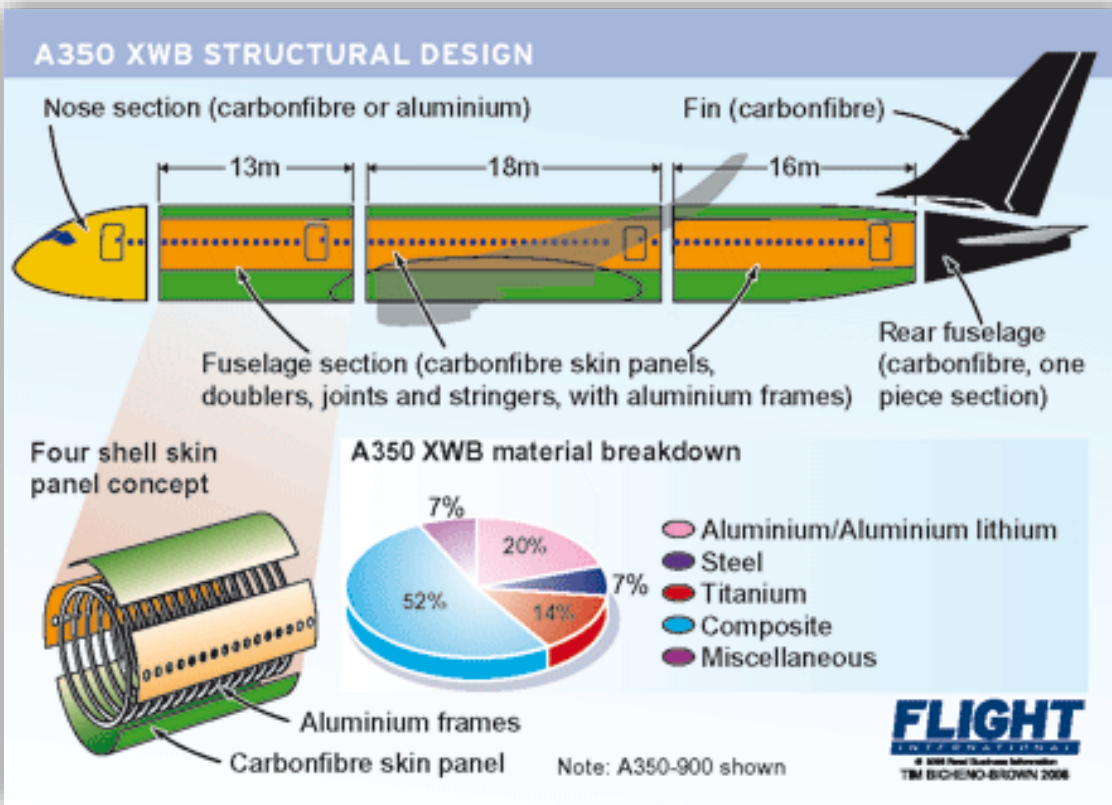


Rear Pressure Bulkhead

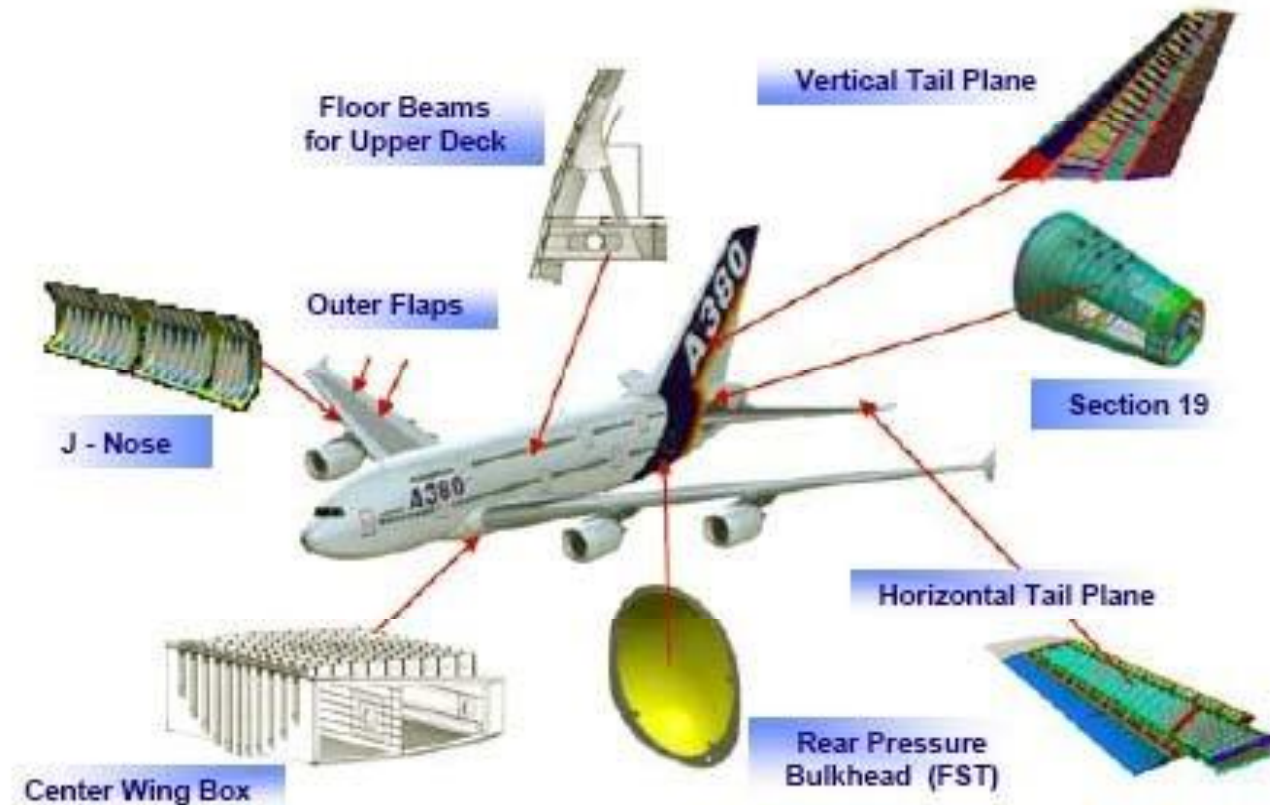
Composite parts of the A340-600

Composites materials in aircraft structures

Advanced composite materials selected for the A350



Composites materials in aircraft structures



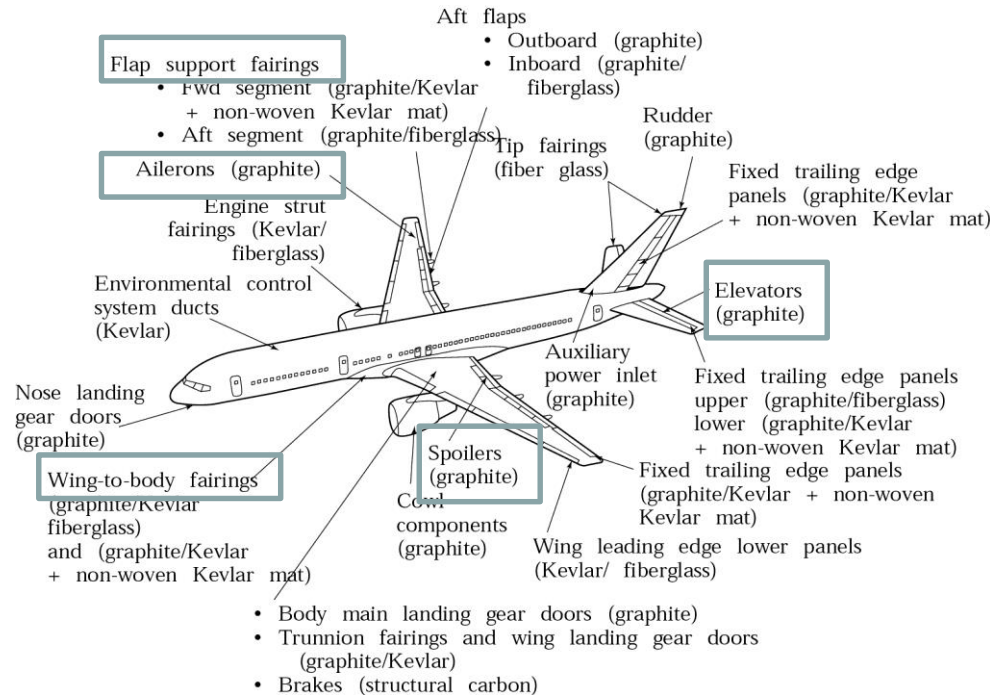
Composite parts of the Airbus 380

A380 New Composite Applications (Examples) (fuente: Karl-Heinz Rendigs. "Airbus and Current Aircrafts Metal Technologies". Materials & Processes. Airbus Germany, Bremen)

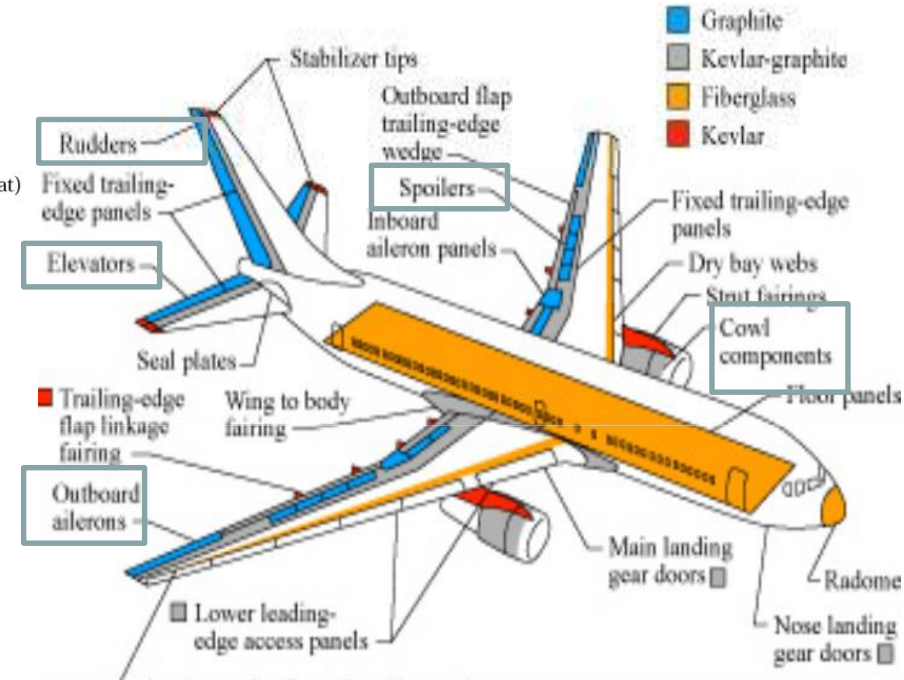
Today: Airbus 380 and A400 M military transport plane includes carbon fibre in **fuselage** or major **parts of the wing**.

Composites materials in aircraft structures

Boeing



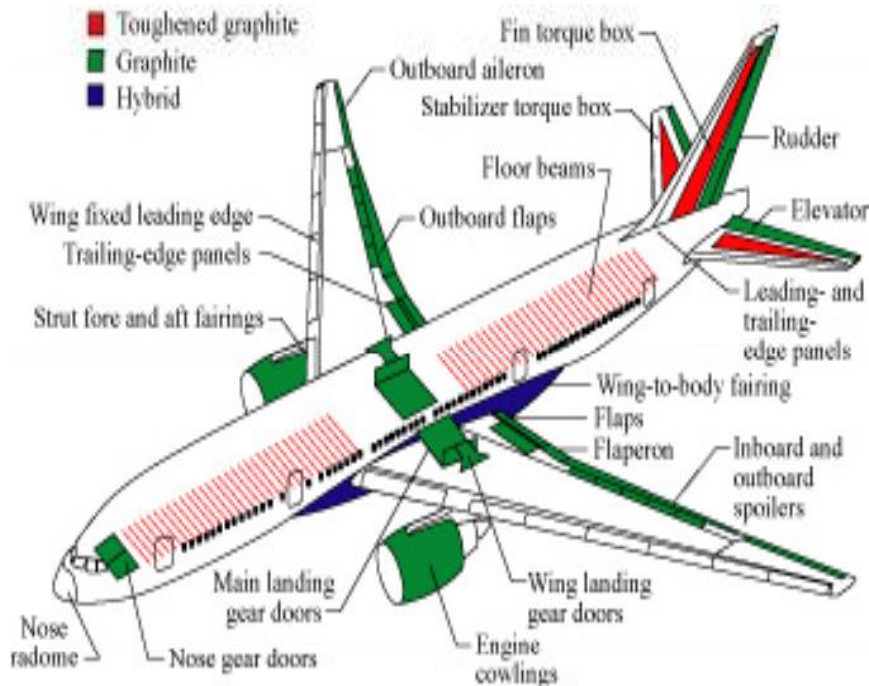
Boeing 757-200 commercial aircraft



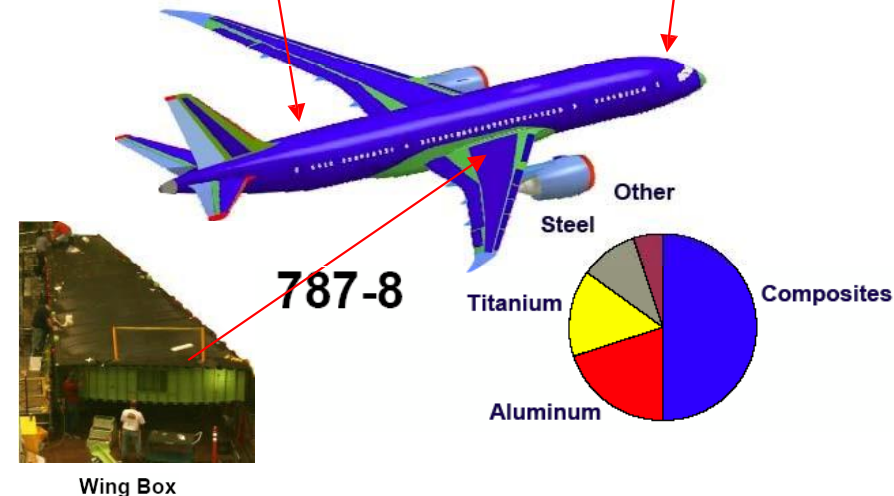
Composite parts of the Boeing 767

Composites materials in aircraft structures

Boeing



Composite parts of the Boeing 777



Boeing 787: use of composite **wing** and **fuselage**. Composite on this aircraft: more than **50% of the structural materials**.

Applications

Typical aircraft fiber composite forms made by the different techniques, as listed

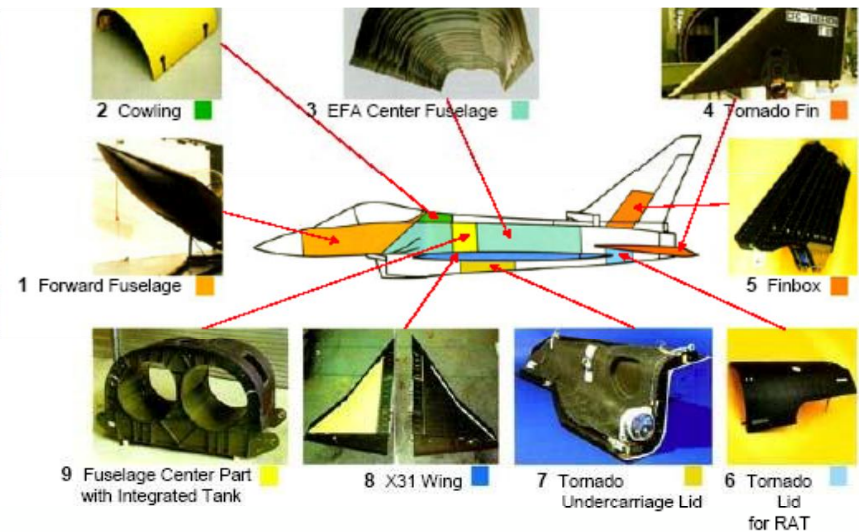
Type of structure	Typical application
<i>Laminates</i>	
Sheets, thick monolithic	Wing skins
Sheets, integrally stiffened	Tail skins
Sandwich panels	Control surfaces, floor sections
Shells	Fuselage sections
Beams	Spars/ribs
Complex forms	Aerofoils
<i>Filament Wound</i>	
Closed shells	Pressure vessels
Open shells	Radomes
	Rocket motors
Tubes	Drive shafts
Secondary formed tubes	Helicopter blades

Type of structure	Typical application
<i>Braided</i>	
Tubes	Drive shafts
Complex tubes	Curved pipes
	Truss joints
	Ducts
Closed shells	Pressure vessels
Secondary formed	Fuselage frames
	Aircraft propellers
	Helicopter blades
<i>Tow placed</i>	
See laminates	See laminates
Complex wraps	Grips
	Shafts
	Ducts
<i>Pultrusion</i>	
Beams	Floor beams
	Stringers
	Spars
	Ribs
	Longerons

Military aircrafts

- Technology programs in the **1970s** were part of **fighter aircraft** development projects such as the F15, Alphajet, Tornado and Mirage. Existing composite knowledge (e.g. moisture pick-up and crack sensitivity) was generated within such programs

Typical composite components for **military aircraft**: fins, undercarriage covers and fuselage components (Tornado, X31 and Eurofighter projects)



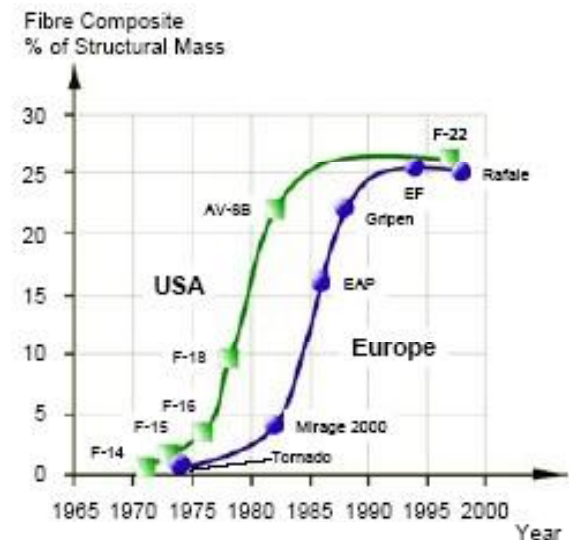
Eurofighter : 70% (surface area) share of composite materials

Military aircrafts

Eurofighter which has **70% (surface area)** share of composite materials..



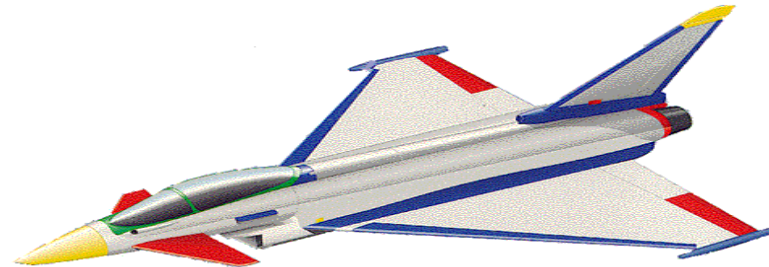
Distribution of materials in Eurofighter



Development of composite share in military aircraft structures in the US and Europe

The variety of materials and technologies in military aircraft is much smaller than that of civil aircraft. **Prepreg technology** still dominates for military applications

Military aircrafts



-  **Carbon fiber composite**
-  **Al/Li alloy**
-  **Ti Alloy**
-  **Glass fiber composite**
-  **Other Al alloys**

USAF B-2 bomber, which is an almost all-composite structure.



Helicopters

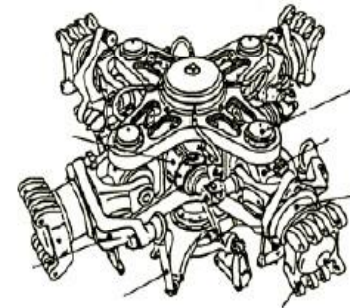
Helicopters composite applications:

- **Rotor blades** (glass fiber reinforced composites)
 - increased in service life by a factor of 200
 - superior fatigue properties
 - potential for multifunctional design
- **Passenger cell**
- **Rotor blade head**
 - Additional functionality: tuned anisotropy
 - Part reduction: integrated hinges
- **Underfloor**
 - Energy absorption capability

Conventional Rotor
Sikorski S 58

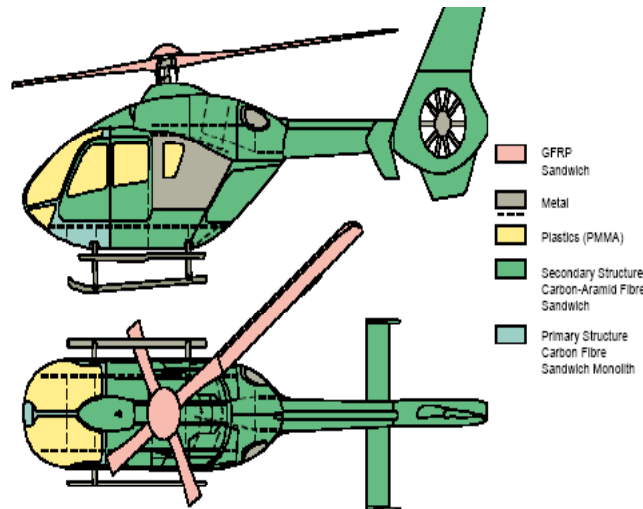


Bearingless Composite Rotor
Eurocopter ATR-Design

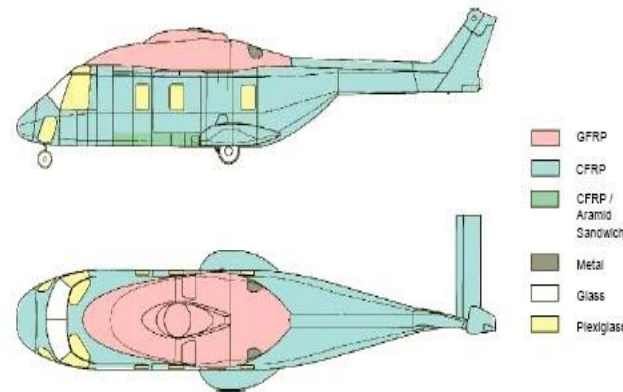


Flexbeam
With Integrated
Control Tube

Rotorblade



Composite parts of the Eurocopter EC135



Composite parts of the Tiger

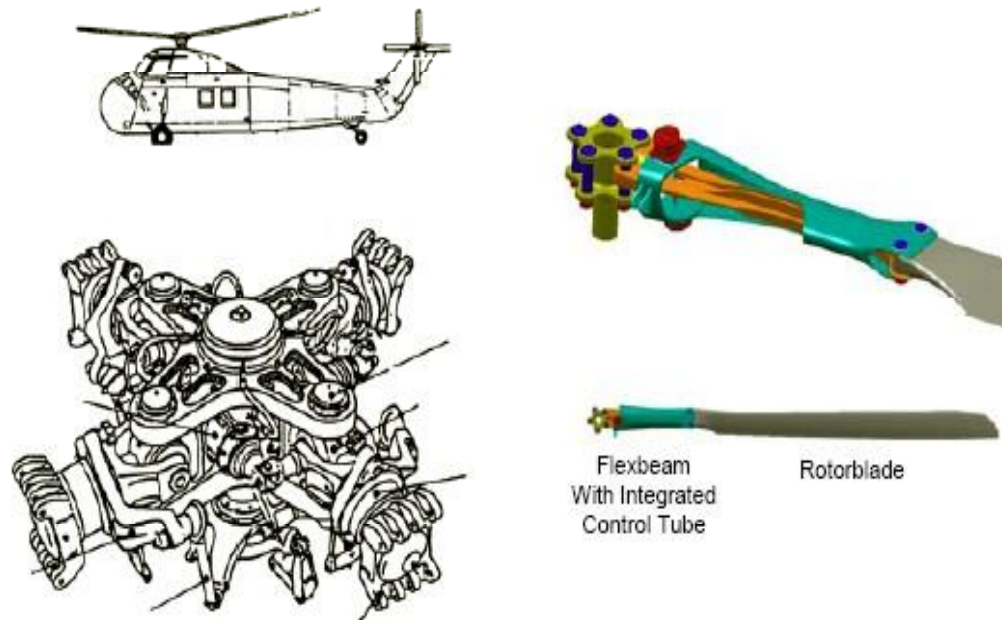
Helicopters

Another specific requirement for helicopters is the *energy absorption capability* of the **underfloor**. Extensive research has been conducted into the development of energy absorbing structures. Due to their high mass specific energy absorption capability, composites are also employed for this purpose.

The **rotor blades** and the **rotor blade head** are typical examples in which composites are used not only for their high specific stiffness and strength, but also for their additional *functionality*. Tuned anisotropy and integrated hinges are just two examples of how composites can be used to improve both performance and cost. The part reduction and integration of the latter leads to a significant reduction of the assembly and maintenance costs, as well as improved reliability

Conventional Rotor
Sikorski S 58

Bearingless Composite Rotor
Eurocopter ATR-Design



Comparison of a conventional and optimized rotor blade head

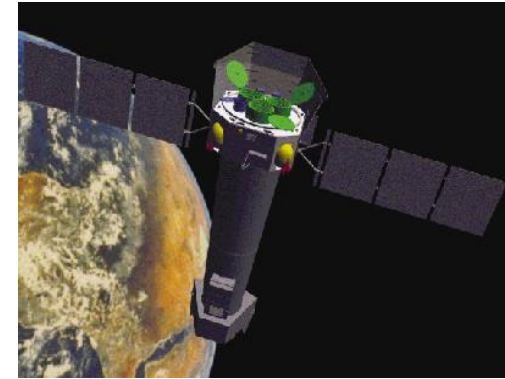
Applications

Aerospace Industry

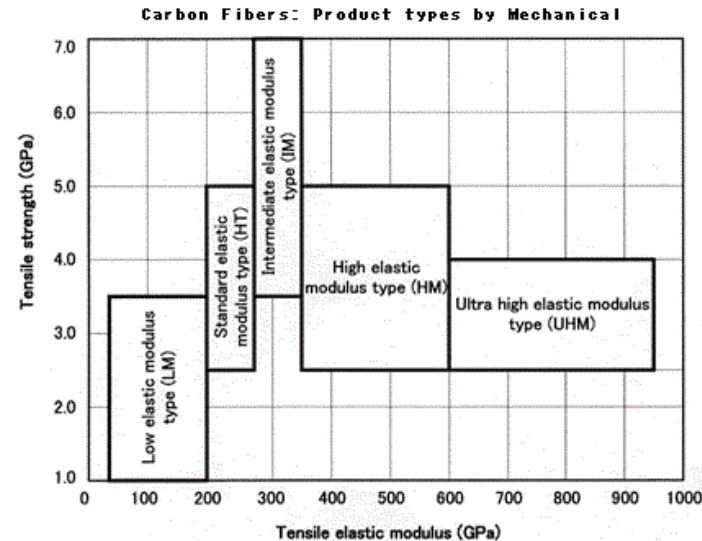
Satellites

- ▶ Structure
- ▶ Antennas
- ▶ Drop-down arms

XMM-Newton

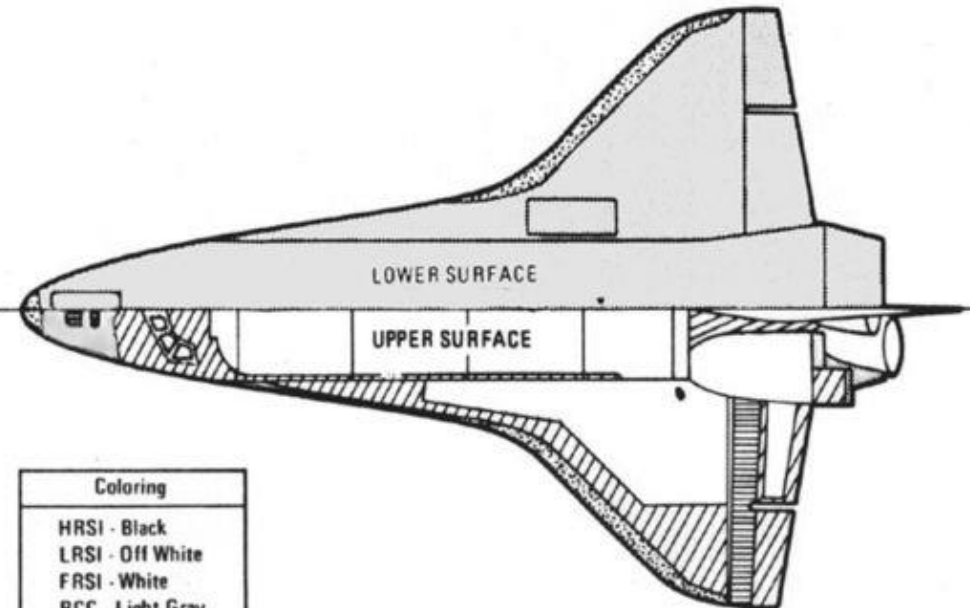
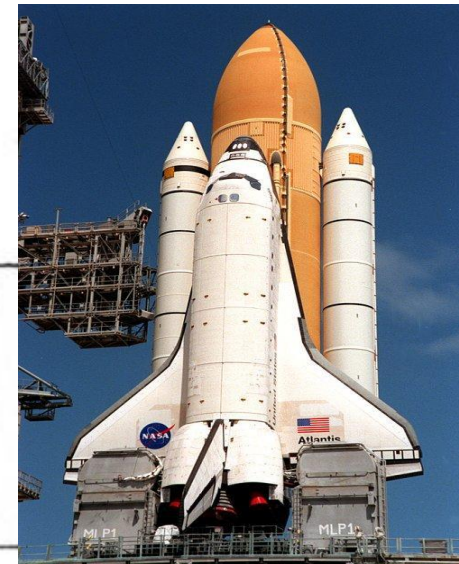
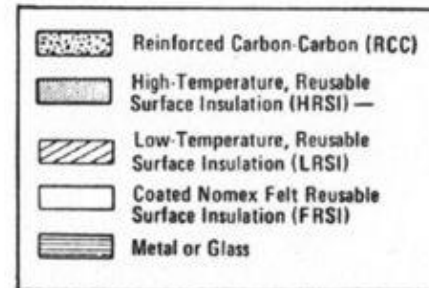
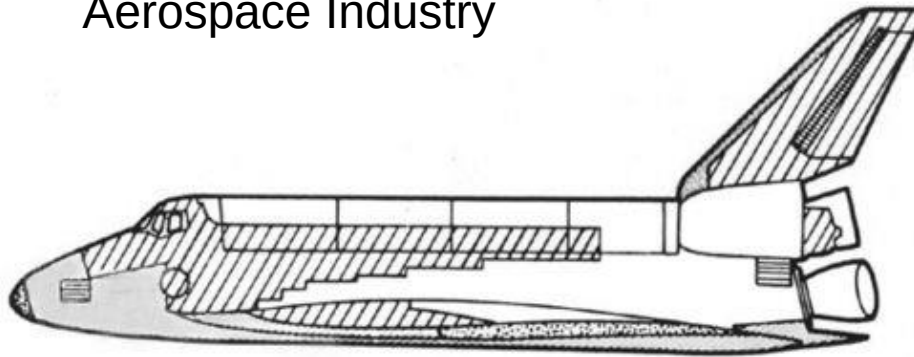


- **Main structural elements:** ultra-high-modulus carbon-fiber composites (low thermal expansion)
- **Telescope tube and mirror support platform:** carbon-fiber with metals composite



Applications

Aerospace Industry



Coloring
HRSI - Black
LRSI - Off White
FRSI - White
RCC - Light Gray

Element*	Area, sq m (sq ft)	Weight, kg (lb)
FRSI	332.7 (3581)	532.1 (1173)
LRSI	254.6 (2741)	1014.2 (2236)
HRSI	479.7 (5164)	4412.6 (9728)
RCC	38.0 (409)	1697.3 (3742)
Miscellaneous		918.5 (2025)
Total	1105.0 (11895)	8574.7 (18,904)

*Includes bulk insulation, thermal barriers, and closeouts

Thermal Protection System, Orbiter 102

Applications

Naval Industry

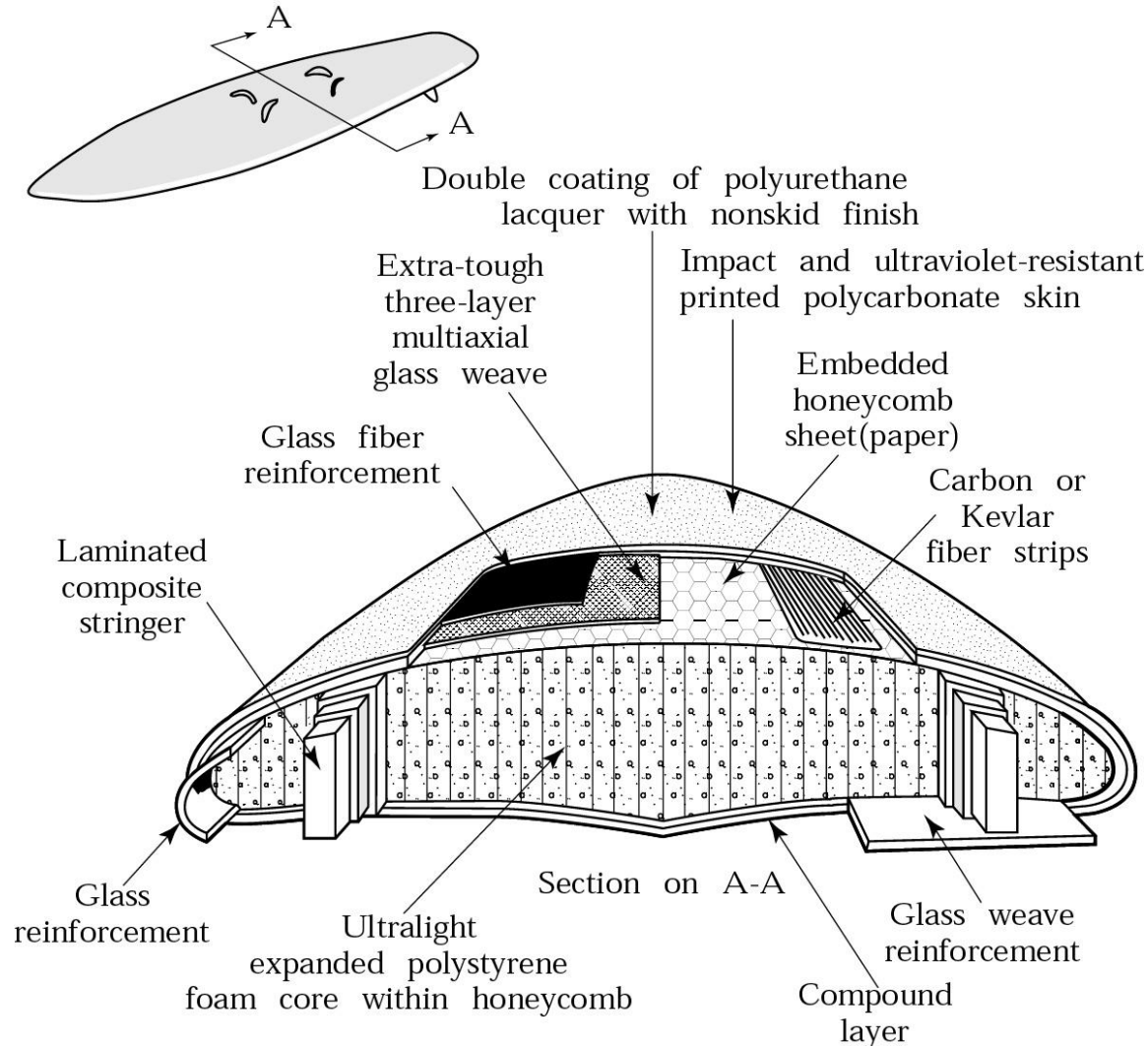
- Tankers
 - Superstructure: (glass/polyester)
 - Hull: (glass/polyester, carbon/epoxy)
 - Mast: (glass/polyester)
- Race boats, sailing boats, yachts
 - Mass application of composites (hull, mast, sail ...)
 - glass/polyester
 - carbon/epoxy
 - laminates are an alternative candidate



Applications

Example of Advanced Materials Construction

Cross-section of a composite sailboard, an example of advanced materials construction



Applications

Automotive Industry



General Motors Ultralite (Manufactured by Scaled Composites)

- **Whole body structure made from composite material**
(Sandwich carbono/epoxi-PVC)

weight: 190 kg

Cost: 13000 \$

6. Applications

Metal-Matrix Composite Materials and Applications

Fiber	Matrix	Applications
Graphite	Aluminum	Satellite, missile, and helicopter structures
	Magnesium	Space and satellite structures
	Lead	Storage-battery plates
	Copper	Electrical contacts and bearings
Boron	Aluminum	Compressor blades and structural supports
	Magnesium	Antenna structures
	Titanium	Jet-engine fan blades
Alumina	Aluminum	Superconductor restraints in fission power reactors
	Lead	Storage-battery plates
	Magnesium	Helicopter transmission structures
Silicon carbide	Aluminum, titanium	High-temperature structures
	Superalloy (cobalt-base)	High-temperature engine components
Molybdenum, tungsten	Superalloy	High-temperature engine components